```
// ===== FILE: Assets/Scripts/Game/GameFlow.cs =====
using System;
using System.Collections;
using System.IO;
using Rescue.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
  /// <summary>启动入口：相机、存档、界面切换。命令行 -shots &lt;dir&gt; 进入自动截图模式。</summary>
  public sealed class GameFlow : MonoBehaviour
  {
    Camera _cam;
    Camera _backdrop;
    HomeScreen _home;
    BattleScreen _battle;
    string _shots;
    int _shotIndex;
    float _aspect;
    [RuntimeInitializeOnLoadMethod(RuntimeInitializeLoadType.AfterSceneLoad)]
    static void Boot()
    {
      if (FindObjectOfType<GameFlow>() != null) return;
      new GameObject("Game").AddComponent<GameFlow>();
    }
    void Awake()
    {
      DontDestroyOnLoad(gameObject);
      Application.targetFrameRate = 60;
      Input.multiTouchEnabled = false;
      _shots = Arg("-shots");
      if (_shots != null) Sfx.Music = false;
      Save.Load();
      if (_shots != null)
      {
        Save.Volatile = true;
        Save.D.tutorial = 0;
      }
      Tween.Init(gameObject);
      Sfx.Init(gameObject);
      _cam = Camera.main;
      if (_cam == null)
      {
        var go = new GameObject("Main Camera");
        go.tag = "MainCamera";
        _cam = go.AddComponent<Camera>();
        go.AddComponent<AudioListener>();
      }
      _cam.orthographic = true;
      _cam.clearFlags = CameraClearFlags.SolidColor;
```

```
51          _cam.backgroundColor = Pal.Paper;
52          _cam.transform.position = new Vector3(0, -0.1f, -10);
53          _cam.nearClipPlane = 0.1f;
54          _cam.farClipPlane = 50f;
55          var back = new GameObject("Backdrop");
56          back.transform.SetParent(transform, false);
57          _backdrop = back.AddComponent<Camera>();
58          _backdrop.orthographic = true;
59          _backdrop.cullingMask = 0;
60          _backdrop.clearFlags = CameraClearFlags.SolidColor;
61          _backdrop.backgroundColor = Pal.Paper;
62          _backdrop.depth = _cam.depth - 1;
63          FitCamera();
64          Ui.Init(transform);
65          if (_shots != null) StartCoroutine(Arg("-ui") != null ? UiShots() : Arg("-fx") != nul...
66          else if (Application.isBatchMode)
67          {
68              Analytics.Begin();
69              ShowHome();
70          }
71          else
72          {
73              CloudSync.Imported += OnCloudImported;
74              var loading = LoadingScreen.Open(transform);
75              CloudSync.Startup(() => StartCoroutine(FinishBoot(loading)));
76          }
77      }
78      void OnDestroy()
79      {
80          CloudSync.Imported -= OnCloudImported;
81      }
82      void OnCloudImported()
83      {
84          if (_battle != null) return;
85          if (_home != null) Destroy(_home.gameObject);
86          _home = null;
87          if (FindObjectOfType<LoadingScreen>() != null) return;
88          ShowHome();
89      }
90      void LateUpdate() => Ui.FitPhone();
91      void Update()
92      {
93          float window = Screen.width / Mathf.Max(1f, Screen.height);
94          if (!Mathf.Approximately(_aspect, window)) FitCamera();
95      }
96      void FitCamera()
97      {
98          _aspect = Screen.width / Mathf.Max(1f, Screen.height);
99          _cam.rect = Ui.CameraViewport();
100         BattleView.FitCamera(_cam);
```

```
        Ui.FitPhone();
        if (_battle != null) _battle.View.CatchLayout();
    }
    IEnumerator FinishBoot(LoadingScreen loading)
    {
        if (loading) yield return loading.WaitMin();
        ShowHome();
        if (!loading) yield break;
        loading.BringToFront();
        yield return loading.FadeOut();
    }
    void ShowHome()
    {
        if (_battle) Destroy(_battle.gameObject);
        _battle = null;
        _home = HomeScreen.Open(transform, StartBattle);
        Sfx.PlayMenu();
    }
    void StartBattle(int chapter) => StartBattle(chapter, false);
    void StartBattle(int chapter, bool autoplay)
    {
        if (_home) Destroy(_home.gameObject);
        _home = null;
        _pending = Resolve(chapter);
        if (BattleScreen.Epic(_pending)) Sfx.PlayEpic();
        else Sfx.PlayStory();
        if (autoplay) OpenBattle(chapter, true);
        else StartCoroutine(MatchThenFight(chapter));
    }
    ChapterDef _pending;
    /// <summary>剧情回直接取；越过终章的序号是聚义厅，按当前阵数和阵容现生成。</summary>
    static ChapterDef Resolve(int chapter)
    {
        if (chapter < ChapterDef.StoryCount) return ChapterDef.All[Mathf.Max(0, chapter)];
        return ChapterDef.Endless(Save.D.endlessRound, Save.Team(), (uint)Environment.TickCou...
    }
    void OpenBattle(int chapter, bool autoplay)
    {
        var loadout = autoplay ? new Loadout() : Save.D.PeekLoadout();
        var ch = _pending ?? Resolve(chapter);
        _pending = null;
        _battle = BattleScreen.Open(transform, ch, _cam, loadout, autoplay, r =>
            ResultPanel.Show(r, () =>
            {
                ShowHome();
                if (autoplay || _home == null) return;
                if (JoinPanel.WantCodex) { JoinPanel.WantCodex = false; _home.ShowCodex(); }
                else _home.ShowShop();
            }));
    }
```

```
151     IEnumerator MatchThenFight(int chapter)
152     {
153         var layer = BuildMatch(chapter);
154         yield return new WaitForSecondsRealtime(1.8f);
155         if (layer) Destroy(layer.gameObject);
156         OpenBattle(chapter, false);
157     }
158     RectTransform BuildMatch(int chapter)
159     {
160         var ch = _pending ?? Resolve(chapter);
161         var layer = Ui.Fill(Ui.Root, Color.white, true, "match");
162         var bg = layer.GetComponent<Image>();
163         bg.sprite = Art.Img("home_bg2");
164         var bleed = layer.Find("bleed")?.GetComponent<Image>();
165         if (bleed != null) bleed.sprite = bg.sprite;
166         Ui.Fill(layer, new Color(0.08f, 0.06f, 0.04f, 0.72f), false, "dim");
167         var title = Ui.Title(layer, "匹配完成", new Vector2(0, 600), 110, Pal.GoldBright);
168         Tween.Pop(title.transform, 0.35f);
169         float mine = Save.D.battles <= 0 ? 0f : (float)Save.D.wins / Save.D.battles;
170         float rivalRate = Mathf.Clamp01(0.36f + ch.RivalSkill * 0.42f);
171         int rivalTier = RankRules.Clamp(Save.D.tier + (ch.RivalSkill >= 0.55f ? 1 : 0) - (ch....
172         int rivalStars = Mathf.RoundToInt(ch.RivalSkill * RankRules.StarsAt(rivalTier));
173         var me = Banner(layer, new Vector2(0, 250), "我", Save.D.tier, Save.D.stars, mine, Ui....
174         var rival = Banner(layer, new Vector2(0, -250), ch.RivalName, rivalTier, rivalStars, ...
175             Pal.Hex("D9452F"), ch.RivalName.Substring(0, 1));
176         SlideIn(me, -1200f);
177         SlideIn(rival, 1200f);
178         var vs = Ui.Icon(layer, "ui_swords", 220, new Vector2(0, 0));
179         var vsText = Ui.Label(layer, "VS", 96, Pal.Hex("FFE36A"), new Vector2(0, -10), new Ve...
180         vsText.fontStyle = FontStyle.Bold;
181         Ui.Stroke(vsText, Ui.Edge, 5f);
182         Tween.Pop(vs.transform, 0.35f, 0.3f);
183         Tween.Pop(vsText.transform, 0.35f, 0.35f);
184         Ui.Label(layer, (ch.IsEndless ? "" : "第" + BattleView.ToCn(ch.Index + 1) + "回 · ") + ...
185         return layer;
186     }
187     static RectTransform Banner(RectTransform layer, Vector2 pos, string name, int tier, int ...
188     {
189         var holder = Ui.Rect(layer, "banner");
190         holder.anchoredPosition = pos;
191         var face = Ui.Framed(holder, new Vector2(900, 250), Vector2.zero, bg, Ui.Edge, 7f, "f...
192         var t = face.transform;
193         Ui.Box(t, new Vector2(860, 30), new Vector2(0, 86), Color.white.A(0.18f), null, "glos...
194         if (glyph == null) Ui.Icon(t, "ui_avatar", 190, new Vector2(-330, 0));
195         else
196         {
197             var av = Ui.Framed(t, new Vector2(176, 176), new Vector2(-330, 0), Pal.Tile, Ui.E...
198             av.pixelsPerUnitMultiplier = 1.2f;
199             Ui.Label(av.transform, glyph, 120, Pal.Ink, new Vector2(0, 6), new Vector2(176, 1...
200         }
```

```
        Ui.Stroke(Ui.Label(t, name, 66, Pal.Paper, new Vector2(-70, 50), new Vector2(400, 90)...
        Ui.Stroke(Ui.Label(t, RankRules.Name(tier), 42, Pal.Hex("FFE36A"), new Vector2(-70, -...
        Ui.RankBadge(t, tier, new Vector2(300, 70), 0.55f);
        var st = Ui.Label(t, Ui.RankStars(tier, stars), 48, Color.white, new Vector2(-70, -72...
        st.supportRichText = true;
        Ui.Stroke(st, Ui.Edge, 2f);
        Ui.Label(t, "胜率", 32, Pal.Paper.A(0.85f), new Vector2(300, -4), new Vector2(200, 50));
        Ui.Stroke(Ui.Label(t, (rate * 100f).ToString("0.0") + "%", 56, Pal.Paper, new Vector2...
        return holder;
    }
    static void SlideIn(RectTransform t, float from)
    {
        var end = t.anchoredPosition;
        t.anchoredPosition = end + new Vector2(from, 0);
        Tween.Run(t, 0.35f, k => { if (t) t.anchoredPosition = end + new Vector2(from * (1f -...
    }
    static string Arg(string name)
    {
        var args = Environment.GetCommandLineArgs();
        for (int i = 0; i < args.Length - 1; i++)
            if (args[i] == name) return args[i + 1];
        return null;
    }
    // ——————————————————————————————————— 自动截图（验收用） ———————————————————————————————————
    IEnumerator Shot(string name)
    {
        yield return new WaitForEndOfFrame();
        Directory.CreateDirectory(_shots);
        ScreenCapture.CaptureScreenshot(Path.Combine(_shots, $"{_shotIndex++:00}_{name}.png"));
        yield return null;
    }
    IEnumerator AutoShots()
    {
        int chapter = int.TryParse(Arg("-chapter"), out var c) ? Mathf.Clamp(c, 0, ChapterDef...
        Save.D.heroes.Add((int)HeroId.WuSong);
        Save.D.heroes.Add((int)HeroId.LinChong);
        ShowHome();
        yield return new WaitForSecondsRealtime(1.2f);
        yield return Shot("home");
        _home.ShowCodex();
        yield return new WaitForSecondsRealtime(0.6f);
        yield return Shot("codex");
        Destroy(_home.gameObject);
        ShowHome();
        _home.ShowShop();
        yield return new WaitForSecondsRealtime(0.6f);
        yield return Shot("shop");
        StartBattle(chapter, true);
        for (int i = 0; i < 2; i++)
        {
```

```
            yield return new WaitForSecondsRealtime(1.6f);
            yield return Shot("scout");
        }
        _battle.Coach("开出来的白地才能摆字。\n招兵偶尔会掉铲子，上方铲子按钮看广告也能领。", 3f);
        yield return new WaitForSecondsRealtime(0.5f);
        yield return Shot("hint");
        int heroShots = 0;
        _battle.View.OnEvent += ev =>
        {
            if (ev.Side == 0 && ev.Type == Ev.HeroFormed && heroShots++ < 2) StartCoroutine(D...
            if (ev.Type == Ev.BossSpawn) StartCoroutine(DelayedShot("boss", 0.6f));
        };
        float t = 0;
        while (_battle != null && !_battle.M.Over && t < 150f)
        {
            yield return new WaitForSecondsRealtime(5f);
            t += 5f;
            if (t % 15f < 0.1f) yield return Shot("battle");
            if (Mathf.Approximately(t, 55f) || Mathf.Approximately(t, 100f))
                for (int i = 0; i < 6; i++)
                {
                    yield return Shot("fx");
                    yield return new WaitForSecondsRealtime(0.06f);
                }
            if (Mathf.Approximately(t, 20f))
            {
                _battle.View.ShowInfoForAnyPiece();
                yield return new WaitForSecondsRealtime(0.4f);
                yield return Shot("info");
            }
            if (Mathf.Approximately(t, 35f))
            {
                _battle.View.ShowShovelOffer();
                yield return new WaitForSecondsRealtime(0.8f);
                yield return Shot("shovelOffer");
                _battle.View.ShovelAd();
                yield return new WaitForSecondsRealtime(1.0f);
                yield return Shot("ad");
                yield return new WaitForSecondsRealtime(3.0f);
                yield return Shot("shovels");
            }
            if (Mathf.Approximately(t, 70f))
            {
                _battle.View.RescueAd();
                yield return new WaitForSecondsRealtime(4.2f);
                yield return Shot("rescue");
            }
        }
        if (_battle != null) _battle.ForceEnd(true);
        yield return new WaitForSecondsRealtime(2.6f);
```

```
301         yield return Shot("result");
302         yield return new WaitForSecondsRealtime(0.5f);
303         Application.Quit();
304     }
305     /// <summary>点选射程对照：-shots DIR -range 1。圆画在我方棋盘中央。</summary>
306     IEnumerator RangeShots()
307     {
308         StartBattle(0, true);
309         yield return new WaitForSecondsRealtime(9f);
310         var view = _battle.View;
311         view.Speed = 0f;
312         var at = new Vector2(0f, BattleView.MyBottom + 3.2f);
313         foreach (UnitKind k in Enum.GetValues(typeof(UnitKind)))
314         {
315             view.PreviewRange(UnitDef.Get(k).Range, at);
316             yield return new WaitForSecondsRealtime(0.6f);
317             yield return Shot("range_" + k);
318         }
319         view.PreviewRange(UnitDef.Get(UnitKind.Gong).Range, new Vector2(BattleView.Left + 0.5...
320         yield return new WaitForSecondsRealtime(0.6f);
321         yield return Shot("range_corner");
322         foreach (var id in new[] { HeroId.WuSong, HeroId.LinChong, HeroId.HuSanNiang, HeroId....
323         {
324             var d = HeroDef.Get(id);
325             view.PreviewRange(d.Range, at, true, d.Disha);
326             yield return new WaitForSecondsRealtime(0.6f);
327             yield return Shot("range_" + id);
328         }
329         Application.Quit();
330     }
331     /// <summary>逐个放出兵种、好汉普攻和大招特效截图：-shots DIR -fx 1。</summary>
332     IEnumerator FxShots()
333     {
334         StartBattle(0, true);
335         yield return new WaitForSecondsRealtime(3.5f);
336         var view = _battle.View;
337         view.Speed = 0f;
338         foreach (UnitKind k in Enum.GetValues(typeof(UnitKind)))
339         {
340             view.ShowcaseUnit(k, 3);
341             yield return new WaitForSecondsRealtime(k == UnitKind.Gong ? 0.1f : 0.07f);
342             yield return Shot("unit_" + k);
343             yield return new WaitForSecondsRealtime(0.5f);
344         }
345         foreach (HeroId id in Enum.GetValues(typeof(HeroId)))
346         {
347             var def = HeroDef.Get(id);
348             if (def.NoAttack) continue;
349             view.ShowcaseHero(id, false);
350             yield return new WaitForSecondsRealtime(0.12f);
```

```
            yield return Shot("atk_" + id);
            yield return new WaitForSecondsRealtime(0.6f);
        }
        view.ShowcaseHero(HeroId.YanQing, true);
        yield return new WaitForSecondsRealtime(0.12f);
        yield return Shot("atk_YanQingCrit");
        yield return new WaitForSecondsRealtime(0.6f);
        foreach (HeroId id in Enum.GetValues(typeof(HeroId)))
        {
            if (HeroDef.Get(id).Hidden) continue;
            view.ShowcaseUlt(id);
            yield return new WaitForSecondsRealtime(0.3f);
            yield return Shot("ult_" + id);
            yield return new WaitForSecondsRealtime(1.2f);
        }
        Application.Quit();
    }
    /// <summary>只截局外界面：-shots DIR -ui 1。</summary>
    IEnumerator UiShots()
    {
        Save.D.coins = 117;
        Save.D.stars = 2;
        Save.D.tier = 13;
        Save.D.claimDay = Save.Today;
        Save.D.claimTier = 12;
        Save.D.claimStars = 3;
        Save.D.claimTaken = false;
        for (int i = 0; i < 25; i++) Save.WinStory(i);
        foreach (var id in Save.D.joined) Save.D.heroes.Add(id);
        Save.D.weapons.AddRange(new[] { (int)HeroId.WuSong, (int)HeroId.WuSong, (int)HeroId.W...
            (int)HeroId.LuZhiShen, (int)HeroId.ZhangHeng, (int)HeroId.SunErNiang });
        Save.D.armed.Add((int)HeroId.WuSong);
        Save.D.armed.Add((int)HeroId.HuaRong);
        ShowHome();
        _home.ShowDaily();
        yield return new WaitForSecondsRealtime(1.2f);
        yield return Shot("daily");
        Destroy(_home.gameObject);
        Save.D.claimTaken = true;
        ShowHome();
        yield return new WaitForSecondsRealtime(1.0f);
        yield return Shot("home");
        Save.D.claimTaken = false;
        yield return Page(() => _home.ShowBoard(), "board");
        yield return Page(() => _home.ShowArmory(), "armory");
        yield return Page(() => { _home.ShowArmory(); _home.SelectWeapon((int)HeroId.WuSong);...
        yield return Page(() => _home.ShowCodex(), "codex");
        yield return Page(() => _home.ShowChapters(), "chapters");
        yield return Page(() => JoinPanel.Show(new System.Collections.Generic.List<HeroId> { ...
        Destroy(Ui.Root.Find("join")?.gameObject);
```

```
401         for (int i = 25; i < ChapterDef.StoryCount; i++) Save.WinStory(i);
402         Save.D.endlessRound = 7;
403         Save.D.endlessBest = 12;
404         yield return Page(() => _home.ShowTeam(), "team");
405         yield return Page(() => JoinPanel.ShowSeats(null), "seats");
406         Destroy(Ui.Root.Find("seats")?.gameObject);
407         Destroy(_home.gameObject);
408         ShowHome();
409         Save.D.coins = 117;
410         _home.ShowShop();
411         yield return new WaitForSecondsRealtime(0.8f);
412         yield return Shot("shop");
413         int first = Save.D.shopOffer[0];
414         Save.D.coins -= ShopItem.All[first].Price;
415         ShopItem.All[first].Give();
416         Save.D.shopSold.Add(first);
417         Save.D.shovels = 2;
418         _home.ShowShop();
419         yield return new WaitForSecondsRealtime(0.8f);
420         yield return Shot("shop_bought");
421         _home.ShowShop(true);
422         yield return new WaitForSecondsRealtime(0.8f);
423         yield return Shot("lotto");
424         _home.DebugSpin();
425         yield return new WaitForSecondsRealtime(3.2f);
426         yield return Shot("lotto_win");
427         Destroy(_home.gameObject);
428         _home = null;
429         var match = BuildMatch(1);
430         yield return new WaitForSecondsRealtime(0.9f);
431         yield return Shot("match");
432         Destroy(match.gameObject);
433         OpenBattle(1, false);
434         yield return new WaitForSecondsRealtime(0.8f);
435         yield return Shot("intro");
436         _battle.ShotItem();
437         yield return new WaitForSecondsRealtime(0.4f);
438         yield return Shot("item");
439         _battle.View.PreviewItemDrag(4, new Vector2(0.6f, BattleView.MyBottom + 2.2f));
440         yield return Shot("item_drag");
441         _battle.View.PreviewItemBubble(0);
442         yield return new WaitForSecondsRealtime(0.3f);
443         yield return Shot("item_bubble");
444         yield return new WaitForSecondsRealtime(1.2f);
445         _battle.ShotPause();
446         yield return new WaitForSecondsRealtime(0.6f);
447         yield return Shot("pause");
448         Destroy(_battle.gameObject);
449         _battle = null;
450         var win = new BattleResult { Chapter = 1, Win = true, Wave = 12, Kills = 230, Rains =...
```

```
            win.Formed.Add(HeroId.WuSong);
            win.Formed.Add(HeroId.ZhangShun);
            win.NewHeroes.Add(HeroId.ZhangShun);
            Save.D.stars = 2;
            ResultPanel.Show(win, null);
            yield return new WaitForSecondsRealtime(2.0f);
            yield return Shot("result_win");
            Destroy(Ui.Root.Find("result").gameObject);
            Save.D.stars = 3;
            ResultPanel.Show(new BattleResult { Chapter = 1, Win = false, Wave = 7, Kills = 96 },...
            yield return new WaitForSecondsRealtime(2.0f);
            yield return Shot("result_lose");
            Destroy(Ui.Root.Find("result").gameObject);
            AdStub.Show("双倍铜钱", null);
            yield return new WaitForSecondsRealtime(0.5f);
            yield return Shot("ad");
            yield return new WaitForSecondsRealtime(1.5f);
            Application.Quit();
        }
        IEnumerator Page(Action open, string name)
        {
            Destroy(_home.gameObject);
            ShowHome();
            yield return null;
            open();
            yield return new WaitForSecondsRealtime(0.8f);
            yield return Shot(name);
        }
        IEnumerator DelayedShot(string name, float delay)
        {
            yield return new WaitForSecondsRealtime(delay);
            yield return Shot(name);
        }
    }
}
// ===== FILE: Assets/Scripts/Game/LoadingScreen.cs =====
using System.Collections;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
    /// <summary>启动页：水墨标题图，底下是健康游戏忠告全文。云存档回来并且最短展示时间到了才进主界面。</summa
    public sealed class LoadingScreen : MonoBehaviour
    {
        public const float MinSeconds = 2.6f;
        const string AdviceTitle = "健康游戏忠告";
        const string Advice =
            "抵制不良游戏，拒绝盗版游戏。\n注意自我保护，谨防受骗上当。\n适度游戏益脑，沉迷游戏伤身。\n合理安排时间，
        RectTransform _root;
        CanvasGroup _group;
```

```
501     float _born;
502     public static LoadingScreen Open(Transform host)
503     {
504         var go = new GameObject("Loading");
505         go.transform.SetParent(host, false);
506         var s = go.AddComponent<LoadingScreen>();
507         s.Build();
508         return s;
509     }
510     void Build()
511     {
512         _born = Time.realtimeSinceStartup;
513         _root = Ui.Rect(Ui.Root, "Loading");
514         _root.anchorMin = Vector2.zero;
515         _root.anchorMax = Vector2.one;
516         _root.offsetMin = _root.offsetMax = Vector2.zero;
517         _group = _root.gameObject.AddComponent<CanvasGroup>();
518         _group.blocksRaycasts = true;
519         var paper = Ui.Rect(_root, "paper");
520         Ui.PinBleed(paper);
521         var paperImg = paper.gameObject.AddComponent<Image>();
522         paperImg.color = Pal.Paper;
523         paperImg.raycastTarget = false;
524         var bg = Ui.Rect(_root, "art");
525         Ui.PinBleed(bg);
526         var img = bg.gameObject.AddComponent<Image>();
527         img.sprite = Splash();
528         img.color = Color.white;
529         img.raycastTarget = true;
530         img.preserveAspect = false;
531         if (img.sprite == null) img.color = Pal.Paper;
532         var title = Ui.Label(_root, AdviceTitle, 36, Pal.Ink, Vector2.zero, new Vector2(920, ...
533             TextAnchor.MiddleCenter, Fonts.Brush, "adviceTitle");
534         var body = Ui.Label(_root, Advice, 30, Pal.Ink, Vector2.zero, new Vector2(980, 176),
535             TextAnchor.UpperCenter, Fonts.Kai, "advice");
536         body.lineSpacing = 1.08f;
537         DockBottom(title.rectTransform, 248, 52);
538         DockBottom(body.rectTransform, 52, 176);
539         _root.SetAsLastSibling();
540     }
541     /// <summary>手写的 meta 若没导成 Sprite，退回整张贴图，避免铺成一块白圆角。</summary>
542     static Sprite Splash()
543     {
544         var sprite = Resources.Load<Sprite>("Art/loading");
545         if (sprite != null && sprite.texture != null && sprite.texture.width > 128) return sp...
546         var tex = Resources.Load<Texture2D>("Art/loading");
547         if (tex == null || tex.width < 128) return null;
548         return Sprite.Create(tex, new Rect(0, 0, tex.width, tex.height), new Vector2(0.5f, 0....
549     }
550     static void DockBottom(RectTransform rt, float y, float h)
```

```
        {
            rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0f);
            rt.pivot = new Vector2(0.5f, 0f);
            rt.sizeDelta = new Vector2(rt.sizeDelta.x, h);
            rt.anchoredPosition = new Vector2(0f, y);
        }
        public void BringToFront()
        {
            if (_root) _root.SetAsLastSibling();
        }
        public IEnumerator WaitMin()
        {
            while (Time.realtimeSinceStartup - _born < MinSeconds) yield return null;
        }
        public IEnumerator FadeOut()
        {
            float t = 0f;
            const float dur = 0.4f;
            while (t < dur)
            {
                t += Time.unscaledDeltaTime;
                if (_group) _group.alpha = 1f - Mathf.Clamp01(t / dur);
                yield return null;
            }
            Destroy(gameObject);
        }
        void OnDestroy()
        {
            if (_root) Destroy(_root.gameObject);
        }
    }
}
// ===== FILE: Assets/Scripts/Game/BattleScreen.cs =====
using System;
using System.Collections.Generic;
using Rescue.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
    public sealed class BattleResult
    {
        public int Chapter;
        /// <summary>聚义厅第几阵；0 是剧情回。</summary>
        public int Round;
        public HeroId[] Pool;
        public bool Win;
        public bool Quit;
        public int Wave, Kills, Rains;
        public List<HeroId> Formed = new List<HeroId>();
```

```
    public List<HeroId> NewHeroes = new List<HeroId>();
}
/// <summary>一局战斗：对局、对手 AI、界面浮层（横幅、引导、暂停、结算）。</summary>
public sealed class BattleScreen : MonoBehaviour
{
    enum Tut { Recruit, Place, Merge, Gold, Hero, Rain, Dismiss, Dig, Danger }
    public Match M { get; private set; }
    public BattleView View { get; private set; }
    public bool Autoplay;
    Action<BattleResult> _done;
    RivalAI _rival, _auto;
    RectTransform _overlay, _bannerLayer, _hint, _intro;
    Text _hintText;
    CanvasGroup _hintGroup;
    bool _hintWant;
    readonly Queue<Action> _banners = new Queue<Action>();
    float _bannerCd;
    bool _started, _ending, _hot;
    float _calm;
    Tut? _tut;
    float _tutT;
    float _ringT;
    float _bossSay;
    Vector2 _ringPos;
    bool _tutorialRun;
    public static BattleScreen Open(Transform host, ChapterDef ch, Camera cam, Loadout items,...
    {
        var go = new GameObject("Battle");
        go.transform.SetParent(host, false);
        var s = go.AddComponent<BattleScreen>();
        s.Autoplay = autoplay;
        s._done = done;
        s.Init(ch, cam, items);
        return s;
    }
    void Init(ChapterDef ch, Camera cam, Loadout items)
    {
        uint seed = (uint)(DateTime.Now.Ticks & 0x7fffffff);
        M = new Match(ch, seed, items);
        M.Hold = true;
        M.Rival.IncomeMul = Mathf.Min(1.12f, ch.RivalIncome + Save.Major * 0.012f);
        float skill = Mathf.Clamp01(ch.RivalSkill + Save.Major * 0.015f);
        _rival = new RivalAI(M.Rival, skill, seed + 99);
        if (Autoplay) _auto = new RivalAI(M.Player, 0.9f, seed + 7);
        View = gameObject.AddComponent<BattleView>();
        View.Build(M, cam);
        View.OnEvent += OnSimEvent;
        View.OnPauseClicked += ShowPause;
        View.OnPlaced += () => { if (_tut == Tut.Place) CompleteTut(); };
        if (Autoplay) { View.InputEnabled = false; View.Speed = 2f; }
```

```
        _overlay = Ui.Rect(Ui.Root, "BattleOverlay");
        _overlay.anchorMin = Vector2.zero;
        _overlay.anchorMax = Vector2.one;
        _overlay.offsetMin = _overlay.offsetMax = Vector2.zero;
        _bannerLayer = Ui.Rect(_overlay, "banners");
        BuildHint();
        _tutorialRun = ch.Index == 0 && !Save.Tut((int)Tut.Hero) && !Autoplay;
        if (Autoplay) Begin();
        else ShowIntro(ch, items);
    }
    void OnDestroy()
    {
        if (_overlay) Destroy(_overlay.gameObject);
    }
    void Begin()
    {
        PlayCalm();
        _started = true;
        M.Hold = false;
        Analytics.LevelStart(M.Chapter.Index + 1, M.Chapter.Name);
        if (_tutorialRun && !Save.Tut((int)Tut.Recruit))
        {
            M.Hold = true;
            StartTut(Tut.Recruit);
        }
    }
    void Update()
    {
        TickHint();
        if (!_started || M.Over || _ending) { UpdateBanners(); return; }
        if (!View.Paused && !M.Hold)
        {
            float dt = Mathf.Min(Time.deltaTime, 0.1f) * View.Speed;
            _rival.Tick(dt);
            _auto?.Tick(dt);
        }
        UpdateTutorial();
        UpdateBanners();
        UpdateMusic();
    }
    /// <summary>头目还在场，或宋江只剩最后一颗心、官兵摸到跟前，就换成「大闹」。险情过去四秒再回到「聚义」，
    void UpdateMusic()
    {
        bool hot = M.Player.InDanger || BossAlive();
        if (hot) { _hot = true; _calm = 0f; }
        else if (_hot) _calm += Time.unscaledDeltaTime;
        if (hot || (_hot && _calm < 4f)) Sfx.PlayCrisis();
        else { _hot = false; PlayCalm(); }
    }
    /// <summary>聚义厅车轮战和终章「蓼儿洼」平时放「百八聚义」，其余各回放「聚义」。</summary>
```

```
    void PlayCalm()
    {
        if (Epic(M.Chapter)) Sfx.PlayEpic();
        else Sfx.PlayBattle();
    }
    public static bool Epic(ChapterDef ch) => ch.IsEndless || ch.Index == ChapterDef.StoryCou...
    bool BossAlive()
    {
        foreach (var s in new[] { M.Player, M.Rival })
            foreach (var e in s.Enemies)
                if (!e.Dead && e.Def.Kind == EnemyKind.Boss) return true;
        return false;
    }
    // ──────────────────────────────────────────────── 开场 ────────────────────────────────────────────
    void ShowIntro(ChapterDef ch, Loadout items)
    {
        var layer = Ui.Fill(_overlay, new Color(0.08f, 0.06f, 0.04f, 0.78f), true, "intro");
        _intro = layer;
        var card = Ui.WoodPanel(Ui.Fit(layer, 1420), new Vector2(940, 1240), new Vector2(0, 4...
        string head = ch.IsEndless ? "聚义厅 · 车轮战" : $"第{BattleView.ToCn(ch.Index + 1)}回 · {ch....
        Ui.Label(card.transform, head, 42, Pal.InkSoft, new Vector2(0, 500), new Vector2(800,...
        Ui.Label(card.transform, ch.Name, 140, Pal.Ink, new Vector2(0, 385), new Vector2(860,...
        if (!string.IsNullOrEmpty(ch.Hui))
            Ui.Label(card.transform, "《水浒传》" + ch.Hui, 32, Pal.Hex("9A6A1E"), new Vector2(0, ...
        Ui.Label(card.transform, ch.Story, 38, Pal.InkSoft, new Vector2(0, 160), new Vector2(...
        var names = new List<string>();
        foreach (var id in ch.Pool) { var d = HeroDef.Get(id); if (!d.Hidden) names.Add(d.Nam...
        Ui.Label(card.transform, "本回好汉：" + string.Join("  ", names), 38, Pal.Gold, new Vector...
        if (ch.Joins.Length > 0)
        {
            var j = new List<string>();
            foreach (var id in ch.Joins) j.Add(HeroDef.Get(id).Nick + HeroDef.Get(id).Name);
            bool done = Save.Joined(ch.Joins[0]);
            Ui.Label(card.transform, (done ? "已入伙：" : "胜后入伙：") + string.Join("、", j), 34, don...
                new Vector2(0, -55), new Vector2(840, 90));
        }
        var nem = HeroDef.Get(ch.Boss.Nemesis).Name;
        Ui.Label(card.transform, $"头目：{ch.Boss.Title}{ch.Boss.Name}（{ch.Boss.SkillText}）\n克星：...
            new Vector2(0, -170), new Vector2(820, 120));
        Ui.Label(card.transform, "对手：" + ch.RivalName + "（谁的宋江先倒谁输）", 32, Pal.InkSoft, new Ve...
        if (items != null && ItemSummary(items).Length > 0)
            Ui.Label(card.transform, "随身：" + ItemSummary(items), 32, Pal.Willow, new Vector2(...
        Ui.CartoonBtn(card.transform, "开  战", new Vector2(480, 140), new Vector2(0, -470), Ui...
        {
            Destroy(layer.gameObject);
            Begin();
            Banner("官军探马", "看清来路，趁官兵未到先征兵布阵", Pal.Gold, 1.1f);
        }, 68);
        Tween.Pop(card.transform, 0.4f);
    }
```

```
    public static string ItemSummary(Loadout l)
    {
        var parts = new List<string>();
        if (l.Gang) parts.Add("生辰纲");
        if (l.Flag) parts.Add("替天行道旗");
        if (l.Marsh) parts.Add("水泊");
        if (l.Stele) parts.Add("石碣");
        if (l.MengHan > 0) parts.Add("蒙汗药×" + l.MengHan);
        if (l.TianShu > 0) parts.Add("天书×" + l.TianShu);
        if (l.Baozi > 0) parts.Add("包子×" + l.Baozi);
        if (l.Xuming) parts.Add("续命丹");
        if (l.Shovels > 0) parts.Add("铲子×" + l.Shovels);
        if (l.Fu > 0) parts.Add("攻速符×" + l.Fu);
        if (l.Trap > 0) parts.Add("陷阱×" + l.Trap);
        if (l.Far) parts.Add("御敌千里");
        return string.Join("、", parts);
    }
    // ─────────────────────────────────────────────── 事件 → 横幅 ──────────────────────────────────────────────
    void OnSimEvent(SimEvent ev)
    {
        bool mine = ev.Side == 0;
        switch (ev.Type)
        {
            case Ev.HeroFormed when mine:
            {
                var d = HeroDef.Get((HeroId)ev.B);
                bool fresh = !Save.D.heroes.Contains((int)d.Id);
                HeroBanner(d, fresh);
                if (_tut == Tut.Gold || _tut == Tut.Hero) CompleteTut();
                if (_tutorialRun) Save.SetTut((int)Tut.Hero);
                break;
            }
            case Ev.WaveStart when ev.A == 1:
                Sfx.Play(Snd.Warn, 0.7f);
                Banner("官兵来了！", "守住宋江，别让他们碰到哥哥", Pal.Red, 0.9f);
                break;
            case Ev.Rain when mine:
                Banner("及时雨！", $"宋江撒钱：炊饼 +{ev.Val:0}，快聚义反推！", Pal.Gold, 1f);
                if (!Save.Tut((int)Tut.Rain)) { Save.SetTut((int)Tut.Rain); ShowHint("宋江每掉一颗心...
                break;
            case Ev.BossSpawn:
            {
                var nem = HeroDef.Get(M.Chapter.Boss.Nemesis).Name;
                Banner(M.Chapter.Boss.Name + " 来了！", $"{M.Chapter.Boss.SkillText} · 克星：{nem}"...
                break;
            }
            case Ev.Bond when mine:
            {
                Bond b = null;
                foreach (var x in Bond.All) if (x.Name == ev.Text) b = x;
```

```
            Sfx.Play(Snd.Join, 0.8f);
            Banner("羁绊 · " + ev.Text, (b != null ? b.Story + "  " : "") + $"伤害 +{Mathf.Rou...
            break;
        }
        case Ev.EliteWarn:
            Banner("捕头来袭", "戴面具的捕头血厚，集中火力", Pal.Hex("3B2050"), 0.8f);
            break;
        case Ev.ItemUsed when mine:
        {
            string sub = ev.Flag == 3 ? "十二秒内全军出手快四成" : ev.Flag == 4 ? "官兵跌进去，一时爬不上来" : "";
            Banner(ev.Text + "！", sub, ev.Flag == 3 ? Pal.Gold : Pal.Willow, 0.9f);
            break;
        }
        case Ev.Ult when ev.Flag == 1:
            if (Time.unscaledTime < _bossSay) break;
            _bossSay = Time.unscaledTime + 7f;
            Banner(ev.Text, M.Chapter.Boss.SkillText, Pal.RedDeep, 0.95f);
            break;
        case Ev.Shovels when mine && ev.Val == 1:
            Banner("方便铲！", "鲁智深送来一把铲子，拖到草地上开荒", Pal.Willow, 0.8f);
            break;
        case Ev.Rescue when mine:
            Banner("宋江哥哥有惊无险！", "公孙胜作法，把官兵都推回路口", Pal.Red, 1.2f);
            break;
        case Ev.Dig when mine && !Save.Tut((int)Tut.Dig):
            Save.SetTut((int)Tut.Dig);
            ShowHint("开出来的白地才能摆字。\n招兵偶尔会掉铲子，上方铲子按钮看广告也能领。", 5f);
            break;
        case Ev.Merge when mine:
            if (_tut == Tut.Merge) CompleteTut();
            break;
        case Ev.Recruit when mine:
            if (_tut == Tut.Recruit) { CompleteTut(); StartTut(Tut.Place); }
            break;
        case Ev.End:
            OnEnd(ev.A == 0);
            break;
    }
}
void HeroBanner(HeroDef d, bool fresh)
{
    Enqueue(() =>
    {
        var layer = Ui.Rect(_bannerLayer, "heroBanner");
        var g = layer.gameObject.AddComponent<CanvasGroup>();
        g.blocksRaycasts = false;
        Ui.Box(layer, new Vector2(1250, 880), new Vector2(0, 180), Pal.Ink.A(0.9f), Art.B...
        Ui.Label(layer, (d.Rank > 0 ? d.Star + " · " : "") + d.Nick, 52, Pal.GoldBright, ...
        var name = Ui.Label(layer, d.Name, 230, Pal.Paper, new Vector2(0, 230), new Vecto...
        Ui.Label(layer, d.Skill, 40, Pal.Paper.A(0.9f), new Vector2(0, 70), new Vector2(9...
```

```
            var seal = Ui.Box(layer, new Vector2(150, 150), new Vector2(330, 320), Pal.Red);
            Ui.Label(seal.transform, "替天\n行道", 44, Pal.Paper, Vector2.zero, new Vector2(150, ...
            seal.rectTransform.localRotation = Quaternion.Euler(0, 0, -12);
            if (fresh) Ui.Label(layer, "— 初次聚义 —", 40, Pal.GoldBright, new Vector2(0, 5), new...
            Ui.Label(layer, "「" + d.Quote + "」", 36, Pal.Paper.A(0.75f), new Vector2(0, -50),...
            Tween.Pop(name.transform, 0.35f);
            seal.transform.localScale = Vector3.one * 3f;
            Tween.Run(seal, 0.25f, k => seal.transform.localScale = Vector3.one * Mathf.Lerp(...
            Tween.FadeGroup(g, 0, 1, 0.15f);
            Tween.FadeGroup(g, 1, 0, 0.35f, () => Destroy(layer.gameObject), 1.9f);
        }, 2.0f);
    }
    void Banner(string title, string sub, Color color, float hold)
    {
        Enqueue(() =>
        {
            var layer = Ui.Rect(_bannerLayer, "banner");
            layer.anchoredPosition = new Vector2(0, 260);
            var g = layer.gameObject.AddComponent<CanvasGroup>();
            g.blocksRaycasts = false;
            Ui.Box(layer, new Vector2(1000, 296), new Vector2(0, -12), Pal.Shadow.A(0.5f), nu...
            Ui.Framed(layer, new Vector2(980, 280), Vector2.zero, Pal.Hex("F6EEDA"), Ui.Edge,...
            Ui.Box(layer, new Vector2(18, 220), new Vector2(-470, 0), color, null, "ribbonL")...
            Ui.Box(layer, new Vector2(18, 220), new Vector2(470, 0), color, null, "ribbonR")....
            int size = Mathf.Min(120, 900 / Mathf.Max(1, title.Length));
            var t = Ui.Label(layer, title, size, color, new Vector2(0, 30), new Vector2(1000,...
            Ui.Stroke(t, Pal.Hex("FFF6E2"), 2.5f);
            if (!string.IsNullOrEmpty(sub)) Ui.Label(layer, sub, 38, Pal.Ink, new Vector2(0, ...
            Tween.Pop(t.transform, 0.3f);
            Tween.FadeGroup(g, 0, 1, 0.12f);
            Tween.FadeGroup(g, 1, 0, 0.3f, () => Destroy(layer.gameObject), hold);
        }, hold * 0.8f + 0.2f);
    }
    void Enqueue(Action show, float duration)
    {
        if (_banners.Count > 3) _banners.Dequeue();
        _banners.Enqueue(() => { show(); _bannerCd = duration; });
    }
    void UpdateBanners()
    {
        _bannerCd -= Time.unscaledDeltaTime;
        if (_bannerCd <= 0 && _banners.Count > 0) _banners.Dequeue()();
    }
    // ——————————————————————————————————————————— 新手引导 ———————————————————————————————————————————
    /// <summary>
    /// 军师小贴士：贴在对手半区顶上（自己的棋盘一格不挡），整块不吃触摸，底下照常拖字；
    /// 点一下贴士就收起，到时也自己淡出。分步引导的「点这/放这」不受影响。
    /// </summary>
    void BuildHint()
    {
```

```
901         _hint = Ui.Rect(_overlay, "hint");
902         Ui.Anchor(_hint, 0.5f, 0.5f);
903         _hint.sizeDelta = new Vector2(HintW, HintH);
904         _hintGroup = _hint.gameObject.AddComponent<CanvasGroup>();
905         _hintGroup.blocksRaycasts = false;
906         _hintGroup.interactable = false;
907         Ui.Box(_hint, new Vector2(HintW, HintH), Vector2.zero, Pal.Ink.A(0.86f));
908         Ui.Box(_hint, new Vector2(96, 96), new Vector2(-HintW * 0.5f + 72, 6), Pal.Red);
909         Ui.Label(_hint, "吴用", 34, Pal.Paper, new Vector2(-HintW * 0.5f + 72, 6), new Vector2(...
910         _hintText = Ui.Label(_hint, "", 37, Pal.Paper, new Vector2(62, 8), new Vector2(HintW ...
911         Ui.Label(_hint, "轻点收起", 25, Pal.Paper.A(0.55f), new Vector2(HintW * 0.5f - 110, -Hint...
912         _hint.gameObject.SetActive(false);
913     }
914     const float HintW = 1000f, HintH = 196f;
915     void ShowHint(string text, float dur)
916     {
917         _hintText.text = text;
918         _hint.gameObject.SetActive(true);
919         _hintWant = true;
920         _hintGroup.alpha = 0f;
921         PlaceHint();
922         Tween.Pop(_hint, 0.25f);
923         _tutT = dur;
924     }
925     public void Coach(string text, float dur) => ShowHint(text, dur);
926     void HideHint(bool now = false)
927     {
928         _hintWant = false;
929         if (now) _hint.gameObject.SetActive(false);
930     }
931     void PlaceHint()
932     {
933         var screen = View.HintScreenTop;
934         if (RectTransformUtility.ScreenPointToLocalPointInRectangle(_overlay, screen, null, o...
935             _hint.anchoredPosition = new Vector2(0f, local.y - HintH * 0.5f);
936     }
937     void TickHint()
938     {
939         if (!_hint.gameObject.activeSelf) return;
940         PlaceHint();
941         if (_hintWant && Input.GetMouseButtonDown(0)
942             && RectTransformUtility.RectangleContainsScreenPoint(_hint, Input.mousePosition, ...
943             _hintWant = false;
944         _hintGroup.alpha = Mathf.MoveTowards(_hintGroup.alpha, _hintWant ? 1f : 0f, Time.unsc...
945         if (!_hintWant && _hintGroup.alpha <= 0f) _hint.gameObject.SetActive(false);
946     }
947     void StartTut(Tut t)
948     {
949         _tut = t;
950         switch (t)
```

```
951         {
952             case Tut.Recruit:
953                 ShowHint("军师吴用：哥哥被押在路尽头了！\n点下面的「聚义」，花炊饼招好汉。", 999);
954                 _ringPos = View.RecruitButtonPos;
955                 break;
956             case Tut.Place:
957                 ShowHint("把山寨里的字拖到路边的白地上，\n他们会自动打路上的官兵。", 999);
958                 _ringPos = BattleView.CampWorld(0);
959                 break;
960             case Tut.Merge:
961                 ShowHint("两个一样、同级的兵拖到一起，\n就能合成升级，更猛！", 7f);
962                 break;
963             case Tut.Gold:
964             {
965                 var names = new List<string>();
966                 foreach (var h in M.Player.Pool) if (!h.Hidden) names.Add(h.Name);
967                 ShowHint("金字是好汉的名字！按顺序摆在一起\n（左→右或上→下）就能召唤：" + string.Join("、", names), 9f);
968                 break;
969             }
970         }
971     }
972     void CompleteTut()
973     {
974         if (_tut == null) return;
975         Save.SetTut((int)_tut.Value);
976         if (_tut == Tut.Place) M.Hold = false;
977         _tut = null;
978         HideHint();
979     }
980     void UpdateTutorial()
981     {
982         if (_tut != null)
983         {
984             _tutT -= Time.deltaTime;
985             if (_tutT <= 0) CompleteTut();
986             if (_tut == Tut.Recruit || _tut == Tut.Place)
987             {
988                 _ringT -= Time.deltaTime;
989                 if (_ringT <= 0)
990                 {
991                     _ringT = 0.9f;
992                     View.FloatText(_ringPos + new Vector2(0, 0.75f), "点这", Pal.Red, 0.4f, 0.8f);
993                     if (_tut == Tut.Place && View.TryFindEmptyCell(out var cell)) View.FloatT...
994                 }
995             }
996         }
997         else if (_hint.gameObject.activeSelf)
998         {
999             _tutT -= Time.deltaTime;
1000            if (_tutT <= 0) HideHint();
```

```
        }
        if (_tut != null || M.Hold || Autoplay) return;
        var p = M.Player;
        if (!Save.Tut((int)Tut.Danger) && p.AtTheGate && !p.RescueUsed)
        {
            Save.SetTut((int)Tut.Danger);
            ShowHint("宋江危险！点山寨右边的「救」，\n看段广告把官兵推回路口。", 4f);
            return;
        }
        if (!Save.Tut((int)Tut.Dismiss) && Save.Tut((int)Tut.Place) && p.EmptyCampSlots == 0 ...
        {
            Save.SetTut((int)Tut.Dismiss);
            ShowHint("山寨满了招不了人？把没用的字\n拖到「聚义」按钮上遣散，还能退炊饼。", 5f);
            return;
        }
        if (!_tutorialRun) return;
        if (!Save.Tut((int)Tut.Merge) && HasMergePair(p)) StartTut(Tut.Merge);
        else if (!Save.Tut((int)Tut.Gold) && HasGoldInCamp(p)) StartTut(Tut.Gold);
    }
    static bool HasMergePair(Side p)
    {
        var all = new List<Piece>();
        foreach (var c in p.Camp) if (c != null) all.Add(c);
        foreach (var c in p.Board.Pieces) if (c != null) all.Add(c);
        for (int i = 0; i < all.Count; i++)
            for (int j = i + 1; j < all.Count; j++)
                if (all[i].Type == PieceType.Unit && Side.CanMerge(all[i], all[j])) return true;
        return false;
    }
    static bool HasGoldInCamp(Side p)
    {
        foreach (var c in p.Camp) if (c != null && c.Type == PieceType.HeroChar) return true;
        return false;
    }
    // ———————————————————————————————— 暂停 / 结束 ————————————————————————————————
    void ShowPause()
    {
        if (M.Over || !_started) return;
        View.Paused = true;
        var layer = Ui.Fill(_overlay, new Color(0, 0, 0, 0.62f), true, "pause");
        var card = Ui.WoodPanel(Ui.Fit(layer, 1200), new Vector2(840, 920), new Vector2(0, 80...
        Ui.Title(card, "暂停", new Vector2(0, 450), 90);
        var gear = Ui.Icon(card, "ui_gear", 96, new Vector2(330, 370));
        var gearState = Ui.Label(card, Sfx.On ? "声音开" : "声音关", 26, Pal.InkSoft, new Vector2(3...
        gearState.fontStyle = FontStyle.Bold;
        Ui.Tap(gear, () =>
        {
            Sfx.On = !Sfx.On;
            Save.D.sfx = Sfx.On;
            Save.Write();
```

```
            if (Sfx.On) Sfx.Play(Snd.Tap);
            gearState.text = Sfx.On ? "声音开" : "声音关";
        });
        string mine = ItemSummary(M.Player.Items);
        int winCoins = ResultPanel.WinCoins(M.Chapter);
        SpoilBox(card, new Vector2(0, 205), "对手战利品： ", mine.Length > 0 ? "你身上的： " + mine : "无");
        SpoilBox(card, new Vector2(0, -45), "战利品： ", $"铜钱 ×{winCoins}  兵器（撑过四波有机会）");
        Ui.CartoonBtn(card, "耻辱撤退", new Vector2(340, 120), new Vector2(-190, -240), Ui.BtnBlu...
        {
            Destroy(layer.gameObject);
            View.Paused = false;
            M.Over = true;
            M.Winner = 1;
            OnEnd(false, true);
        }, 54);
        Ui.Label(card, "投降没有战利品", 28, Pal.InkSoft, new Vector2(-190, -330), new Vector2(340, ...
        Ui.CartoonBtn(card, "死战不休", new Vector2(340, 120), new Vector2(190, -240), Ui.BtnRed,...
        Ui.Label(card, "胜利者拿走所有战利品", 28, Pal.InkSoft, new Vector2(190, -330), new Vector2(360...
        Tween.Pop(card, 0.25f);
    }
    static void SpoilBox(RectTransform card, Vector2 pos, string title, string body)
    {
        var box = Ui.Box(card, new Vector2(720, 210), pos, Pal.Hex("E2D2AE"), null, "spoil");
        box.pixelsPerUnitMultiplier = 1f;
        Ui.Stroke(Ui.Label(card, title, 40, Pal.Hex("D9822B"), pos + new Vector2(0, 62), new ...
        Ui.Label(card, body, 34, Pal.Ink, pos + new Vector2(0, -24), new Vector2(680, 110));
    }
    /// <summary>验收截图用：跳过开场，塞一张攻速符和一个陷阱，立刻用掉攻速符。</summary>
    public void ShotItem()
    {
        if (_intro) Destroy(_intro.gameObject);
        if (!_started) Begin();
        M.Player.Items.Fu = 1;
        M.Player.Items.Trap = 1;
        M.Player.Items.MengHan = 2;
        M.Player.UseFu();
    }
    /// <summary>验收截图用：跳过开场直接打开暂停面板。</summary>
    public void ShotPause()
    {
        if (_intro) Destroy(_intro.gameObject);
        if (!_started) Begin();
        ShowPause();
    }
    /// <summary>验收截图用：直接判定结局。</summary>
    public void ForceEnd(bool win)
    {
        if (M.Over) return;
        M.Over = true;
        M.Winner = win ? 0 : 1;
```

```
            OnEnd(win);
        }
        void OnEnd(bool win, bool quit = false)
        {
            if (_ending) return;
            _ending = true;
            HideHint(true);
            if (win) Sfx.PlayVictory();
            else Sfx.PlayDefeat();
            var r = new BattleResult
            {
                Chapter = M.Chapter.Index, Round = M.Chapter.Round, Pool = M.Chapter.Pool, Win = ...
            };
            foreach (var id in M.Player.FormedEver)
            {
                r.Formed.Add(id);
                if (!Save.D.heroes.Contains((int)id)) r.NewHeroes.Add(id);
            }
            if (quit) Analytics.LevelFail("quit", M.Wave, M.Player.Kills);
            else if (win) Analytics.LevelClear(M.Wave, M.Player.Kills);
            else Analytics.LevelFail("lose", M.Wave, M.Player.Kills);
            Banner(win ? "对手的宋江倒了！" : "宋江被押走了", win ? "这一阵，赢了" : "再来！", win ? Pal.Gold : Pal.Ink,...
            Tween.Run(this, 1.6f, k => { }, () => _done?.Invoke(r));
        }
    }
}
// ===== FILE: Assets/Scripts/Game/BattleView.cs =====
using System;
using System.Collections.Generic;
using Rescue.Sim;
using UnityEngine;
namespace Rescue
{
    /// <summary>战场的世界空间表现与输入。只读 Match 状态渲染，玩家操作走 Side 的公开接口。</summary>
    public sealed partial class BattleView : MonoBehaviour
    {
        public const float Left = -4f;
        /// <summary>9:16 时自己棋盘的底边。更高的屏幕整块下移，见 LowerShift。</summary>
        public const float MyBottom0 = -4.25f;
        /// <summary>比 9:16 多出来的高度。下半场（自己的棋盘、山寨、聚义）整体下移这么多，底边留白和 9:16 一样。</su
        public static float LowerShift { get; private set; }
        public static float MyBottom => MyBottom0 - LowerShift;
        public const float OppBottom = 1.13f;
        public static float CampY => MyBottom - 0.47f;
        public static float ButtonY => CampY - 1.22f;
        /// <summary>被动道具一行，排在聚义下面。主动道具和聚义同一行，靠左。</summary>
        public static float ItemY => ButtonY - 1.06f;
        public const float TopY = 6.39f;
        static float WallY => (MyBottom + Board.Rows + OppBottom) * 0.5f;
        const float HudHalf = 0.40f;
```

```
    const float ItemGap = 1.0f;
    float _activeScale = 0.96f;
    float _activeGap = 1.15f;
    /// <summary>可见宽度正好是 8 格。再为了装高度而缩小，两侧就会露出地面。</summary>
    const float WidthNeed = Board.Cols;
    public static Vector3 CamBase = new Vector3(0f, -0.1f, -10f);
    const int MaxShovelAds = 1;
    const float RecruitW = 2.7f;
    const float RecruitH = 1.0f;
    public Match M;
    public bool Paused;
    public float Speed = 1f;
    public bool InputEnabled = true;
    public event Action<SimEvent> OnEvent;
    public event Action OnPauseClicked;
    public event Action<Piece> OnDragStart;
    public event Action OnPlaced;
    Camera _cam;
    Transform _root, _fxRoot, _low, _frameRoot;
    float _layoutShift;
    SpriteRenderer _midInk, _midPaper;
    float _acc;
    float _shake;
    int _frame;
    readonly Transform[][,] _cellVis = { new Transform[Board.Cols, Board.Rows], new Transform...
    readonly Cell[][,] _cellKind = { new Cell[Board.Cols, Board.Rows], new Cell[Board.Cols, B...
    readonly SpriteRenderer[,] _hl = new SpriteRenderer[Board.Cols, Board.Rows];
    readonly TextMesh[,] _hlMark = new TextMesh[Board.Cols, Board.Rows];
    readonly Dictionary<int, PieceView> _pieces = new Dictionary<int, PieceView>();
    readonly Dictionary<int, EnemyView> _enemies = new Dictionary<int, EnemyView>();
    readonly Dictionary<int, FrameView> _frames = new Dictionary<int, FrameView>();
    readonly GoalView[] _goals = new GoalView[2];
    readonly List<Fx> _fx = new List<Fx>();
    readonly SpriteRenderer[] _campSlots = new SpriteRenderer[Tuning.CampSlots];
    SpriteRenderer _recruitBtn, _recruitFill, _pauseBtn, _mengBtn, _shuBtn, _baoziBtn, _fuBtn...
    readonly CooldownRing[] _itemCd = new CooldownRing[5];
    readonly TextMesh[] _itemCdText = new TextMesh[5];
    readonly bool[] _itemCooling = new bool[5];
    readonly float[] _itemFlash = new float[5];
    readonly List<SpriteRenderer> _passiveBtns = new List<SpriteRenderer>();
    readonly List<ShopItem> _passiveItems = new List<ShopItem>();
    TextMesh _recruitCost, _recruitLabel, _bunsText, _waveText, _midText, _mengText, _shuText...
    Transform _shovelRoot;
    float _shovelSlide;
    bool _shovelForce;
    const float ShovelOfferX = 3.47f;
    const int ShovelOfferEmpty = 2;
    /// <summary>白地比格子小一圈，圆角，格与格之间留出草地。边上和中间同一块。</summary>
    const float TilePad = 0.88f;
    Transform _bunIcon, _rescueRoot;
```

```
    static readonly Vector3 _bunIconScale = new Vector3(0.5f, 0.5f, 1f);
    int _shovelAds;
    TextMesh _tipText;
    SpriteRenderer _tipBg;
    float _tipT;
    float _recruitShown;
    float _recruitFlash;
    bool _recruitWasFull;
    // 点选属性卡 + 攻击范围。卡贴在山寨下面，不压棋盘，射程圈才能看全。
    Transform _card;
    SpriteRenderer _cardBg, _cardShadow, _cardPaper, _cardSeal, _cardRule, _cardIcon;
    TextMesh _cardTitle, _cardLevel, _cardBody, _cardGlyph;
    Vector2 _cardSize;
    float _cardT;
    int _cardFrame;
    string _cardKey;
    PieceView _picked;
    // 闲置提示
    readonly List<SpriteRenderer> _hintDots = new List<SpriteRenderer>();
    SpriteRenderer _hintBox;
    float _idleT, _hintRecalc;
    bool _hintOn;
    Vector2 _hintFrom, _hintTo;
    // 拖拽状态
    PieceView _drag;
    int _dragCamp = -1;
    CellPos _dragCell;
    Vector2 _downPos;
    bool _dragMoved;
    readonly TextMesh[] _campPlus = new TextMesh[Tuning.CampSlots];
    readonly List<SpriteRenderer> _dragDots = new List<SpriteRenderer>();
    bool _rowRefresh;
    // ———————————————————————————————— 布局 ————————————————————————————————
    /// <summary>
    /// 宽度永远刚好是 8 格，左右贴边。不够高就收紧界面，不再把整张棋盘缩小。
    /// 真机上标题让开微信胶囊。比 9:16 更高时，多出来的高度补在两军之间，下半场贴着屏幕底，底下不再空出一大块。
    /// </summary>
    public static void FitCamera(Camera cam)
    {
        var rect = cam.rect;
        float vw = Mathf.Max(1f, rect.width * Screen.width), vh = Mathf.Max(1f, rect.height *...
        float bar = 0f, menuTop = 0f, menuBottom = 0f, bottomPx = 0f;
        if (rect.width >= 0.999f && rect.height >= 0.999f)
            Ui.ScreenInsets(out bar, out menuTop, out menuBottom, out _, out bottomPx);
        float upp = WidthNeed / vw;
        float topWorld = menuBottom > 0f
            ? TopY + (menuTop + menuBottom) * 0.5f * upp
            : TopY + HudHalf + 0.02f + bar * upp;
        cam.orthographicSize = vh * upp * 0.5f;
        CamBase = new Vector3(0f, topWorld - cam.orthographicSize, -10f);
```

```
1251        cam.transform.position = CamBase;
1252        float viewH = cam.orthographicSize * 2f;
1253        float designH = WidthNeed * (Ui.DesignH / Ui.DesignW);
1254        float extra = Mathf.Max(0f, viewH - designH);
1255        // 底部安全区只留指示条那一截。报得太大时不能把多出来的高度又吃回去。
1256        float inset = Mathf.Clamp(bottomPx, 0f, vh * 0.08f) * upp;
1257        LowerShift = Mathf.Max(0f, extra - inset);
1258    }
1259    public static Vector2 World(int side, V2 p) =>
1260        side == 0 ? new Vector2(Left + p.x, MyBottom + p.y) : new Vector2(Left + (Board.Cols ...
1261    public static Vector2 CellWorld(int side, int c, int r) => World(side, new CellPos(c, r)....
1262    public static Vector2 CampWorld(int i) => new Vector2(-2.3f + i * 1.12f, CampY);
1263    public Vector2 BunsWorld => _bunIcon.position;
1264    public void Build(Match m, Camera cam)
1265    {
1266        M = m;
1267        _cam = cam;
1268        _root = new GameObject("World").transform;
1269        _root.SetParent(transform, false);
1270        _low = new GameObject("Low").transform;
1271        _low.SetParent(_root, false);
1272        _layoutShift = LowerShift;
1273        _fxRoot = new GameObject("Fx").transform;
1274        _fxRoot.SetParent(transform, false);
1275        var paper = Gfx.Sprite(_root, Art.Paper, Pal.Paper, new Vector2(0, 0), new Vector2(30...
1276        paper.drawMode = SpriteDrawMode.Tiled;
1277        paper.size = new Vector2(30, 30);
1278        paper.transform.localScale = Vector3.one;
1279        BuildFrame();
1280        for (int s = 0; s < 2; s++) BuildBoard(s);
1281        BuildScouts();
1282        BuildCamp();
1283        BuildHud();
1284        BuildTip();
1285        BuildCard();
1286        BuildActives();
1287        BuildBubble();
1288        for (int s = 0; s < 2; s++) _goals[s] = new GoalView(this, s);
1289        BuildAmbient();
1290        SyncAll(true);
1291    }
1292    /// <summary>屏幕变高后，把已经摆好的下半场平移到新位置。父节点停在原点，本地坐标仍是世界坐标。</summary
1293    public void CatchLayout()
1294    {
1295        if (_low == null) return;
1296        float dy = -(LowerShift - _layoutShift);
1297        if (Mathf.Abs(dy) < 0.001f) return;
1298        _layoutShift = LowerShift;
1299        for (int i = 0; i < _low.childCount; i++)
1300        {
```

```
            var c = _low.GetChild(i);
            var p = c.localPosition;
            p.y += dy;
            c.localPosition = p;
        }
        foreach (var v in _pieces.Values)
            if (v.Side == 0) v.Target.y += dy;
        if (_card != null && _card.gameObject.activeSelf)
        {
            var p = _card.localPosition;
            p.y += dy;
            _card.localPosition = p;
        }
        if (_bubble != null && _bubble.gameObject.activeSelf)
        {
            var p = _bubble.localPosition;
            p.y += dy;
            _bubble.localPosition = p;
        }
        BuildFrame();
        PlaceMid();
    }
    /// <summary>原作是一整张棋盘：上半对手、下半自己，中间一道城墙。更高的屏幕把多出来的空补成草地。</summa
    void BuildFrame()
    {
        if (_frameRoot != null) Destroy(_frameRoot.gameObject);
        _frameRoot = new GameObject("frame").transform;
        _frameRoot.SetParent(_root, false);
        float gapBottom = MyBottom + Board.Rows;
        float gapTop = OppBottom;
        float gapH = Mathf.Max(0.05f, gapTop - gapBottom);
        float mid = (gapBottom + gapTop) * 0.5f;
        if (gapH > 0.5f)
        {
            int row = 0;
            for (float y = gapBottom + 0.5f; y < gapTop + 0.51f; y += 1f, row++)
                for (int c = -2; c < Board.Cols + 2; c++)
                    Gfx.Sprite(_frameRoot, Art.Ground("tex_grass", c, row), Color.white, new ...
        }
        float stoneH = Mathf.Min(0.38f, gapH);
        var wall = Gfx.Sprite(_frameRoot, Art.Img("tex_stone"), Color.white, new Vector2(0, m...
        wall.drawMode = SpriteDrawMode.Tiled;
        wall.transform.localScale = Vector3.one;
        wall.size = new Vector2(Board.Cols, stoneH);
        Gfx.Sprite(_frameRoot, Art.White, Pal.Ink.A(0.75f), new Vector2(0, mid + stoneH * 0.5...
        Gfx.Sprite(_frameRoot, Art.White, Pal.Ink.A(0.75f), new Vector2(0, mid - stoneH * 0.5...
        for (float x = Left + 0.25f; x < -Left; x += 0.5f)
        {
            Gfx.Sprite(_frameRoot, Art.White, Pal.InkSoft.A(0.85f), new Vector2(x, mid + ston...
            Gfx.Sprite(_frameRoot, Art.White, Pal.InkSoft.A(0.85f), new Vector2(x, mid - ston...
```

```
1351        }
1352        DecorateWall(mid, stoneH);
1353    }
1354    void PlaceMid()
1355    {
1356        if (_midText == null) return;
1357        float y = WallY;
1358        if (_midInk != null) _midInk.transform.localPosition = new Vector2(0f, y);
1359        if (_midPaper != null) _midPaper.transform.localPosition = new Vector2(0f, y);
1360        _midText.transform.localPosition = new Vector2(0f, y);
1361    }
1362    void BuildBoard(int side)
1363    {
1364        var b = M.Sides[side].Board;
1365        for (int c = 0; c < Board.Cols; c++)
1366            for (int r = 0; r < Board.Rows; r++)
1367            {
1368                BuildCell(side, c, r);
1369                if (side == 0)
1370                {
1371                    var pos = CellWorld(side, c, r);
1372                    _hl[c, r] = Gfx.Sprite(_low, Art.Rounded, Color.clear, pos, new Vector2(0...
1373                    _hlMark[c, r] = Gfx.Text(_low, "", Fonts.Kai, 0.26f, Pal.RedDeep, pos + n...
1374                }
1375            }
1376        var sp = CellWorld(side, b.Spawn.c, b.Spawn.r);
1377        BuildSpawn(side == 0 ? _low : _root, sp);
1378    }
1379    void BuildCell(int side, int c, int r)
1380    {
1381        var b = M.Sides[side].Board;
1382        var kind = b.Cells[c, r];
1383        _cellKind[side][c, r] = kind;
1384        if (_cellVis[side][c, r] != null) Destroy(_cellVis[side][c, r].gameObject);
1385        var t = new GameObject("cell").transform;
1386        t.SetParent(side == 0 ? _low : _root, false);
1387        t.localPosition = CellWorld(side, c, r);
1388        _cellVis[side][c, r] = t;
1389        var full = new Vector2(1.01f, 1.01f);
1390        switch (kind)
1391        {
1392            case Cell.Path:
1393                Gfx.Sprite(t, Art.Ground("tex_earth", c, r), Color.white, Vector2.zero, full,...
1394                PathEdges(t, side, b, c, r);
1395                break;
1396            case Cell.Grass:
1397            case Cell.Marsh:
1398                Gfx.Sprite(t, Art.Ground("tex_grass", c, r), Color.white, Vector2.zero, full,...
1399                break;
1400            case Cell.Tile:
```

```
            Gfx.Sprite(t, Art.Ground("tex_grass", c, r), Color.white, Vector2.zero, full,...
            Gfx.Sprite(t, Art.Rounded, Pal.Shadow, new Vector2(0.02f, -0.03f), new Vector...
            Gfx.Sprite(t, Art.Rounded, Pal.Tile, Vector2.zero, new Vector2(TilePad, TileP...
            break;
        case Cell.Water:
            Gfx.Sprite(t, Art.White, Pal.Water, Vector2.zero, full, 8);
            for (int i = 0; i < 3; i++)
                Gfx.Sprite(t, Art.Stroke, Color.white.A(0.5f), new Vector2(0, 0.22f - i *...
            break;
        case Cell.Rock:
            Gfx.Sprite(t, Art.Ground("tex_grass", c, r), Color.white, Vector2.zero, full,...
            break;
    }
    Decorate(t, side, kind, c, r);
    BleedEdge(t, side, c, r, kind);
}
/// <summary>最外侧一列的地面向外铺出去，万一画面略宽，两侧仍是地面而不是露底。</summary>
void BleedEdge(Transform t, int side, int c, int r, Cell kind)
{
    if (c != 0 && c != Board.Cols - 1) return;
    int outward = (side == 0 ? 1 : -1) * (c == 0 ? -1 : 1);
    var sprite = kind == Cell.Water ? Art.White : Art.Ground(kind == Cell.Path ? "tex_ear...
    var color = kind == Cell.Water ? Pal.Water : Color.white;
    for (int i = 1; i <= 3; i++)
        Gfx.Sprite(t, sprite, color, new Vector2(outward * i, 0f), new Vector2(1.01f, 1.0...
}
void PathEdges(Transform t, int side, Board b, int c, int r)
{
    var self = CellWorld(side, c, r);
    int salt = 0;
    foreach (var d in new[] { new CellPos(1, 0), new CellPos(-1, 0), new CellPos(0, 1), n...
    {
        salt++;
        int nc = c + d.c, nr = r + d.r;
        if (Board.InBounds(nc, nr) && b.Cells[nc, nr] == Cell.Path) continue;
        var delta = (CellWorld(side, nc, nr) - self) * 0.5f;
        var edge = Gfx.Sprite(t, Art.Stroke, Pal.PathEdge.A(0.8f), delta * 0.94f, new Vec...
        edge.transform.localRotation = Quaternion.Euler(0, 0, Mathf.Abs(delta.x) > 0.01f ...
        PathPebbles(t, delta, side, c, r, salt);
    }
}
void BuildCamp()
{
    Gfx.Sprite(_low, Art.Rounded, Pal.Ink.A(0.85f), new Vector2(-0.52f, CampY), new Vecto...
    Gfx.Sprite(_low, Art.Rounded, Pal.PaperDark, new Vector2(-0.52f, CampY), new Vector2(...
    Gfx.Text(_low, "山寨", Fonts.Brush, 0.4f, Pal.RedDeep, new Vector2(-3.43f, CampY), 5);
    for (int i = 0; i < _campSlots.Length; i++)
    {
        _campSlots[i] = Gfx.Sprite(_low, Art.Rounded, Pal.Paper.A(0.9f), CampWorld(i), ne...
        _campPlus[i] = Gfx.Text(_low, "", Fonts.Kai, 0.48f, Pal.InkSoft, CampWorld(i), 6);
```

```
1451        }
1452        _rescueRoot = new GameObject("rescue").transform;
1453        _rescueRoot.SetParent(_low, false);
1454        _rescueRoot.localPosition = new Vector2(ShovelOfferX, CampY);
1455        Gfx.Sprite(_rescueRoot, Art.Rounded, Pal.Shadow, new Vector2(0.03f, -0.07f), new Vect...
1456        _rescueBtn = Gfx.Sprite(_rescueRoot, Art.Rounded, Pal.GoldBright, Vector2.zero, new V...
1457        Gfx.Sprite(_rescueRoot, Art.Frame, Pal.Gold, Vector2.zero, new Vector2(0.84f, 1.08f),...
1458        Gfx.Text(_rescueRoot, "救", Fonts.Brush, 0.5f, Pal.RedDeep, new Vector2(0, 0.12f), 52);
1459        Gfx.Sprite(_rescueRoot, Art.Rounded, Pal.Red, new Vector2(0, -0.34f), new Vector2(0.7...
1460        Gfx.Sprite(_rescueRoot, Art.Img("ui_video"), Color.white, new Vector2(-0.2f, -0.34f),...
1461        Gfx.Text(_rescueRoot, "广告", Fonts.Kai, 0.17f, Pal.Paper, new Vector2(0.12f, -0.34f), ...
1462        _rescueRoot.gameObject.SetActive(false);
1463        _shovelRoot = new GameObject("shovelOffer").transform;
1464        _shovelRoot.SetParent(_low, false);
1465        _shovelRoot.localPosition = new Vector2(ShovelOfferX, CampY);
1466        Gfx.Sprite(_shovelRoot, Art.Rounded, Pal.Shadow, new Vector2(0.03f, -0.07f), new Vect...
1467        Gfx.Sprite(_shovelRoot, Art.Rounded, Pal.Hex("E89A2C"), Vector2.zero, new Vector2(0.8...
1468        Gfx.Sprite(_shovelRoot, Art.Rounded, Pal.Hex("FFC94A"), new Vector2(0, 0.03f), new Ve...
1469        Gfx.Sprite(_shovelRoot, Art.Img("shovel"), Color.white, new Vector2(0.04f, -0.02f), V...
1470        Gfx.Sprite(_shovelRoot, Art.Img("ui_video"), Color.white, new Vector2(-0.24f, 0.27f),...
1471        Gfx.Text(_shovelRoot, "x" + Tuning.AdShovels, Fonts.Kai, 0.26f, Pal.Ink, new Vector2(...
1472        Gfx.Text(_shovelRoot, "x" + Tuning.AdShovels, Fonts.Kai, 0.26f, Color.white, new Vect...
1473        _shovelRoot.gameObject.SetActive(false);
1474    }
1475    /// <summary>白地快用完时挂在山寨右边，一直等到玩家看掉；手里有铲子或要救命时先收起来。</summary>
1476    void TickShovelOffer(float dt, bool rescue)
1477    {
1478        var p = M.Player;
1479        bool want = _shovelAds < MaxShovelAds && !M.Over && !rescue && (_shovelForce || Shove...
1480        bool shown = _shovelRoot.gameObject.activeSelf;
1481        if (!want)
1482        {
1483            if (shown) _shovelRoot.gameObject.SetActive(false);
1484            return;
1485        }
1486        if (!shown) SlideInShovel();
1487        _shovelSlide = Mathf.Min(1f, _shovelSlide + dt / 0.35f);
1488        float x = Mathf.Lerp(ShovelOfferX + 1.4f, ShovelOfferX, Tween.OutBack(_shovelSlide));
1489        float wiggle = Mathf.Sin(Time.time * 9f) * 7f * (Mathf.Sin(Time.time * 1.6f) > 0.4f ?...
1490        _shovelRoot.localPosition = new Vector2(x, CampY + Mathf.Sin(Time.time * 3f) * 0.03f);
1491        _shovelRoot.localRotation = Quaternion.Euler(0, 0, wiggle);
1492    }
1493    static bool HasShovel(Side p)
1494    {
1495        if (p.ShovelStock > 0) return true;
1496        foreach (var q in p.Camp) if (q != null && q.Type == PieceType.Shovel) return true;
1497        return false;
1498    }
1499    /// <summary>白地最多还剩两格、还有草地能开、手里没铲子。</summary>
1500    static bool ShovelMoment(Side p)
```

```
12752         var content = Rect(view, "content");
12753         content.anchorMin = new Vector2(0.5f, 1f);
12754         content.anchorMax = new Vector2(0.5f, 1f);
12755         content.pivot = new Vector2(0.5f, 1f);
12756         content.sizeDelta = new Vector2(size.x, Mathf.Max(size.y, contentHeight));
12757         content.anchoredPosition = Vector2.zero;
12758         var sr = view.gameObject.AddComponent<ScrollRect>();
12759         sr.content = content;
12760         sr.horizontal = false;
12761         sr.vertical = true;
12762         sr.movementType = ScrollRect.MovementType.Elastic;
12763         sr.scrollSensitivity = 40f;
12764         sr.decelerationRate = 0.12f;
12765         view.gameObject.AddComponent<ScrollHold>().Init(sr);
12766         return content;
12767     }
12768     /// <summary>把滚动区停在离顶 y 处（弹出动画期间也保持住）。</summary>
12769     public static void ScrollTo(RectTransform content, float y)
12770     {
12771         var hold = content.parent.GetComponent<ScrollHold>();
12772         float max = Mathf.Max(0, content.sizeDelta.y - ((RectTransform)content.parent).sizeDe...
12773         y = Mathf.Clamp(y, 0, max);
12774         content.anchoredPosition = new Vector2(0, y);
12775         if (hold) hold.Y = y;
12776     }
12777     /// <summary>content 里按“离顶多远”摆子物体（子物体锚点在 content 中心）。</summary>
12778     public static Vector2 Top(RectTransform content, float x, float yFromTop) => new Vector2(...
12779 }
12780 /// <summary>
12781 /// 面板弹出时父节点从零缩放，ScrollRect 会按退化的边界把内容弹走；
12782 /// 开头这一小段时间里钉住起始位置，玩家一碰就放手。
12783 /// </summary>
12784 public sealed class ScrollHold : MonoBehaviour
12785 {
12786     public float Y;
12787     ScrollRect _sr;
12788     float _t = 0.6f;
12789     public void Init(ScrollRect sr) => _sr = sr;
12790     void LateUpdate()
12791     {
12792         if (_sr == null || _t <= 0) return;
12793         _t -= Time.unscaledDeltaTime;
12794         if (Input.GetMouseButton(0) || Input.touchCount > 0) { _t = 0; return; }
12795         _sr.StopMovement();
12796         _sr.content.anchoredPosition = new Vector2(0, Y);
12797     }
12798 }
12799 /// <summary>极简补间：不依赖第三方库，用不受时间缩放影响的时钟。</summary>
12800 public sealed class Tween : MonoBehaviour
12801 {
```

```
12802     sealed class Job
12803     {
12804         public Func<float, bool> Step;
12805         public float T, Dur, Delay;
12806         public Action Done;
12807         public UnityEngine.Object Owner;
12808     }
12809     static Tween _i;
12810     readonly List<Job> _jobs = new List<Job>();
12811     public static void Init(GameObject host) { if (_i == null) _i = host.AddComponent<Tween>(...
12812     public static void Run(UnityEngine.Object owner, float dur, Action<float> step, Action do...
12813     {
12814         _i._jobs.Add(new Job { Owner = owner, Dur = Mathf.Max(0.0001f, dur), Delay = delay, D...
12815     }
12816     void Update()
12817     {
12818         float dt = Time.unscaledDeltaTime;
12819         for (int i = _jobs.Count - 1; i >= 0; i--)
12820         {
12821             var j = _jobs[i];
12822             if (j.Owner == null) { _jobs.RemoveAt(i); continue; }
12823             if (j.Delay > 0) { j.Delay -= dt; continue; }
12824             j.T += dt;
12825             float k = Mathf.Clamp01(j.T / j.Dur);
12826             j.Step(k);
12827             if (k >= 1f)
12828             {
12829                 _jobs.RemoveAt(i);
12830                 j.Done?.Invoke();
12831             }
12832         }
12833     }
12834     public static float OutBack(float k) { const float s = 1.70158f; k -= 1; return k * k * (...
12835     public static float OutCubic(float k) { k -= 1; return k * k * k + 1; }
12836     public static void Pop(Transform t, float dur = 0.35f, float delay = 0)
12837     {
12838         t.localScale = Vector3.zero;
12839         Run(t, dur, k => t.localScale = Vector3.one * OutBack(k), null, delay);
12840     }
12841     public static void Punch(Transform t, float dur = 0.2f, float amount = 0.15f, Vector3? re...
12842     {
12843         var baseScale = restScale ?? Vector3.one;
12844         Run(t, dur, k => t.localScale = baseScale * (1f + Mathf.Sin(k * Mathf.PI) * amount), ...
12845     }
12846     public static void FadeGroup(CanvasGroup g, float from, float to, float dur, Action done ...
12847     {
12848         g.alpha = from;
12849         Run(g, dur, k => g.alpha = Mathf.Lerp(from, to, k), done, delay);
12850     }
12851 }
```

```
}
// ===== FILE: Assets/Scripts/Sim/Chapters.cs =====
using System;
using System.Collections.Generic;
namespace Rescue.Sim
{
    public enum BossSkill { Summon, Haste, Resist }
    public sealed class BossDef
    {
        public string Name;
        public string Title;
        public BossSkill Skill;
        public HeroId Nemesis;
        public string SkillText;
    }
    /// <summary>Grass/Marsh 未开垦不能摆；Tile/Water 是开好的地。</summary>
    public enum Cell { Grass, Path, Water, Tile, Rock, Marsh }
    /// <summary>一回战斗。剧情回按原著顺序排开，赢下后 Joins 里的好汉入伙；通关后是聚义厅车轮战。</summary>
    public sealed class ChapterDef
    {
        public int Index;
        public string Name;
        /// <summary>原著回目。</summary>
        public string Hui;
        /// <summary>所属篇章。</summary>
        public string Act;
        public string Story;
        /// <summary>水浒小课堂：赢下后给玩家讲的一段原著典故。</summary>
        public string Lore;
        public string[] Map;
        public HeroId[] Pool;
        public HeroId[] Joins = new HeroId[0];
        public BossDef Boss;
        public float RivalSkill;
        public float RivalIncome = 1f;
        /// <summary>保留字段。双方官兵血量相同，不再给对手单独加厚。</summary>
        public float RivalFoeHp = 1f;
        /// <summary>这一回双方官兵的血量。第一回是 1，越往后两边一起变厚。</summary>
        public float FoeHp = 1f;
        public string RivalName;
        /// <summary>聚义厅第几阵；0 表示剧情回。</summary>
        public int Round;
        public bool IsEndless => Round > 0;
        // ——————————————————————————————————————————————— 地图模板 ———————————————————————————————————————————————
        public static readonly Dictionary<string, string[]> Maps = new Dictionary<string, string[]>
        {
            ["qingfeng"] = new[] { "S######.", ".ooo..#.", ".######.", ".#ooo...", ".######G" },
            ["jiangzhou"] = new[] { "S.######", "#.Go.o.#", "#o..o..#", "#.o...o#", "########" },
            ["bailong"] = new[] { "~~~~~~~S", "########", "#.ooooo.", "########", ".....G" },
            ["snake"] = new[] { "S#######", ".oooo..#", "..######", "..#oooo.", "..#####G" },
```

```
12902       ["ring"] = new[] { "S#######", "...oo..#", ".o....o#", "...oo..#", "G#######" },
12903       ["jiuqu"] = new[] { "S.###.#G", "#o#.#.#.", "#.#o#o#.", "#o#.#.#o", "###.###." },
12904       ["marsh"] = new[] { "~~~~S###", "~ww..o.#", "########", "#~~o~~w.", "#######G" },
12905       ["zigzag"] = new[] { "S####...", ".oo.#o..", "....####", "..o.o.o#", "G#######" },
12906       ["plain"] = new[] { "S#######", ".o..o..#", "G.o..o.#", "#.o.o..#", "########" },
12907       ["river"] = new[] { "S###.###", ".o.#o#o#", ".ww#o#~#", "~..###w#", "..~....G" },
12908   };
12909   static readonly string[] EndlessMaps = { "qingfeng", "jiangzhou", "bailong", "snake", "ri...
12910   // ──────────────────────────────────────────────────────── 剧情 ────────────────────────────────────────────────────────
12911   public static readonly ChapterDef[] All = BuildStory();
12912   public static int StoryCount => All.Length;
12913   static ChapterDef C(string name, string hui, string act, string map, HeroId[] joins, Hero...
12914       string boss, string title, BossSkill skill, HeroId nemesis, string rival, string stor...
12915   {
12916       var pool = new List<HeroId>(joins);
12917       foreach (var h in extra) if (!pool.Contains(h)) pool.Add(h);
12918       return new ChapterDef
12919       {
12920           Name = name, Hui = hui, Act = act, Map = Maps[map], Joins = joins, Pool = pool.To...
12921           Boss = new BossDef { Name = boss, Title = title, Skill = skill, Nemesis = nemesis...
12922           RivalName = rival, Story = story, Lore = lore,
12923       };
12924   }
12925   static string SkillText(BossSkill s, HeroId nemesis) => s switch
12926   {
12927       BossSkill.Summon => "每隔几秒大喝来人，招来三个兵丁",
12928       BossSkill.Haste => "喝令周围兵丁加速猛冲",
12929       _ => $"老奸巨猾，只受一半伤害（{HeroDef.Get(nemesis).Name}例外）",
12930   };
12931   static HeroId[] J(params HeroId[] ids) => ids;
12932   static ChapterDef[] BuildStory()
12933   {
12934       var list = new[]
12935       {
12936           C("史家村", "第二回 王教头私走延安府 九纹龙大闹史家村", "引首", "snake",
12937               J(HeroId.ShiJin, HeroId.ZhuWu, HeroId.ChenDa, HeroId.YangChun), J(),
12938               "县尉", "华阴县", BossSkill.Summon, HeroId.ShiJin, "王进",
12939               "少华山的跳涧虎陈达要借道华阴县，被九纹龙史进活捉。朱武、杨春负荆请罪，史进敬他们义气，结为兄弟——
12940               "《水浒传》开篇，洪太尉误走妖魔，从伏魔殿放出一百单八个魔君，正是日后梁山的三十六天罡、七十二地煞。"
12941           C("拳打镇关西", "第三回 史大郎夜走华阴县 鲁提辖拳打镇关西", "鲁达篇", "qingfeng",
12942               J(HeroId.LuZhiShen, HeroId.LiZhong), J(HeroId.ShiJin, HeroId.ZhuWu),
12943               "郑屠", "镇关西", BossSkill.Haste, HeroId.LuZhiShen, "金老汉",
12944               "渭州状元桥下，卖肉的郑屠自称镇关西，欺压卖唱的金家父女。提辖鲁达听了，拍案而起：洒家这就去会会他！"
12945               "鲁达三拳：一拳打在鼻子上，咸的酸的辣的一发滚出；二拳打在眼眶，红的黑的绛的都绽将出来；三拳太阳穴，
12946           C("桃花村", "第五回 小霸王醉入销金帐 花和尚大闹桃花村", "鲁达篇", "jiangzhou",
12947               J(HeroId.ZhouTong), J(HeroId.LuZhiShen, HeroId.LiZhong, HeroId.ShiJin),
12948               "崔道成", "生铁佛", BossSkill.Resist, HeroId.ShiJin, "刘太公",
12949               "鲁智深大闹五台山被打发去东京，路过桃花村，醉打要强娶民女的小霸王周通；转到瓦罐寺，又撞上恶僧生铁佛
12950               "鲁智深醉打周通之后，周通与李忠占的桃花山，后来也成了梁山的一支人马。"),
12951           C("野猪林", "第八回 林教头刺配沧州道 鲁智深大闹野猪林", "林冲篇", "ring",
```

```
12952       J(HeroId.LinChong), J(HeroId.LuZhiShen, HeroId.ShiJin, HeroId.LiZhong),
12953       "薛霸", "押解公人", BossSkill.Haste, HeroId.LuZhiShen, "董超",
12954       "林冲被高俅陷害，刺配沧州。押解公人董超、薛霸受了贿赂，要在野猪林结果他性命，水火棍刚举起——一条禅杖
12955       "董超、薛霸一路用滚水烫林冲的脚，鲁智深一路暗中护送，这才有了野猪林救林冲。"),
12956   C("柴进庄", "第九回 柴进门招天下客 林冲棒打洪教头", "林冲篇", "plain",
12957       J(HeroId.ChaiJin), J(HeroId.LinChong, HeroId.LuZhiShen),
12958       "洪教头", "柴府教师", BossSkill.Resist, HeroId.LinChong, "庄客",
12959       "沧州小旋风柴进专爱结交好汉，盛情款待林冲。庄上的洪教头不服，硬要比棒——林冲只得出手。",
12960       "柴进是后周世宗柴荣的后人。宋太祖陈桥兵变得了柴家天下，所以赐了丹书铁券，犯罪也不得加刑。"),
12961   C("风雪山神庙", "第十回 林教头风雪山神庙 陆虞候火烧草料场", "林冲篇", "zigzag",
12962       J(HeroId.ZhuGui), J(HeroId.LinChong, HeroId.ChaiJin, HeroId.LuZhiShen),
12963       "陆谦", "陆虞候", BossSkill.Haste, HeroId.LinChong, "富安",
12964       "大雪压塌了草屋，林冲到山神庙避风，听见门外陆谦放火烧了草料场，要他葬身火海。忍无可忍，林冲挺枪而出。
12965       "林冲被逼得家破人亡，只得投奔梁山——这就是成语「逼上梁山」的来历。"),
12966   C("杨志卖刀", "第十二回 梁山泊林冲落草 汴京城杨志卖刀", "杨志篇", "river",
12967       J(HeroId.YangZhi), J(HeroId.LinChong, HeroId.ZhuGui, HeroId.ChaiJin),
12968       "牛二", "没毛大虫", BossSkill.Summon, HeroId.YangZhi, "王伦",
12969       "杨家将后人杨志失陷花石纲，穷得只能在汴京街头卖祖传宝刀。泼皮牛二偏要他当街试刀杀人……",
12970       "杨志的宝刀有三好：砍铜剁铁刀口不卷，吹毛得过，杀人刀上没血。"),
12971   C("智取生辰纲", "第十六回 杨志押送金银担 吴用智取生辰纲", "生辰纲篇", "ring",
12972       J(HeroId.WuYong, HeroId.GongSunSheng, HeroId.LiuTang, HeroId.BaiSheng), J(Her...
12973       "老都管", "梁府都管", BossSkill.Summon, HeroId.BaiSheng, "梁中书",
12974       "梁中书给岳父蔡京送十万贯生辰纲。黄泥冈上烈日当头，一个卖酒的汉子挑着担子唱着山歌走来……",
12975       "白胜唱的「赤日炎炎似火烧，野田禾稻半枯焦」，是书中最有名的一首山歌。"),
12976   C("二龙山", "第十七回 花和尚单打二龙山 青面兽双夺宝珠寺", "生辰纲篇", "jiangzhou",
12977       J(HeroId.CaoZheng), J(HeroId.YangZhi, HeroId.LuZhiShen, HeroId.LiZhong),
12978       "邓龙", "二龙山寨主", BossSkill.Resist, HeroId.YangZhi, "周通",
12979       "丢了生辰纲，杨志无路可走，在曹正酒店里遇见了鲁智深。曹正献计：绑了花和尚诈作献俘，打上二龙山！",
12980       "二龙山宝珠寺后来成为三山聚义中的一山，鲁智深、杨志、武松、施恩、张青、孙二娘都在这里。"),
12981   C("私放晁天王", "第十八回 美髯公智稳插翅虎 宋公明私放晁天王", "生辰纲篇", "snake",
12982       J(HeroId.ZhuTong, HeroId.LeiHeng), J(HeroId.WuYong, HeroId.LiuTang, HeroId.Go...
12983       "府尹", "济州", BossSkill.Summon, HeroId.ZhuTong, "何涛",
12984       "生辰纲案发，济州官兵连夜来捉晁盖。押司宋江飞马报信，都头朱仝、雷横一前一后，都想放晁天王一条生路。"
12985       "宋江私放晁盖，是整个梁山故事的转折点——没有这一放，就没有后来的梁山泊。"),
12986   C("石碣村", "第十九回 晁盖梁山小夺泊", "生辰纲篇", "marsh",
12987       J(HeroId.RuanXiaoEr, HeroId.RuanXiaoWu, HeroId.RuanXiaoQi), J(HeroId.WuYong, ...
12988       "何涛", "缉捕观察", BossSkill.Summon, HeroId.RuanXiaoQi, "黄安",
12989       "缉捕观察何涛带着官船杀进石碣村芦苇荡。阮家三兄弟笑了：在水上和打鱼的斗？",
12990       "吴用去石碣村说三阮入伙，三兄弟拍着脖子说：「这腔热血，只要卖与识货的！」"),
12991   C("火并王伦", "第十九回 林冲水寨大并火", "生辰纲篇", "snake",
12992       J(HeroId.DuQian, HeroId.SongWan), J(HeroId.LinChong, HeroId.WuYong, HeroId.Zh...
12993       "王伦", "白衣秀士", BossSkill.Resist, HeroId.LinChong, "晁盖",
12994       "晁盖一伙投奔梁山，寨主王伦却拿出五锭大银要打发他们下山。林冲踢翻桌子：你这嫉贤妒能的贼！",
12995       "王伦心胸狭窄、嫉贤妒能。林冲火并王伦，拥晁盖为寨主，梁山从此兴旺起来。"),
12996   C("怒杀阎婆惜", "第二十二回 阎婆大闹郓城县 朱仝义释宋公明", "宋江篇", "jiuqu",
12997       J(HeroId.SongQing), J(HeroId.ZhuTong, HeroId.LeiHeng, HeroId.ChaiJin),
12998       "张文远", "郓城押司", BossSkill.Haste, HeroId.ZhuTong, "阎婆",
12999       "阎婆惜拿到晁盖写给宋江的书信，要挟宋江。一时失手，宋江杀了她。张文远唆使阎婆告官，朱仝又一次放走了宋
13000       "「及时雨」是说宋江扶危济困，像及时雨一样帮人。他因此名满江湖，好汉们一听名号就纳头便拜。"),
13001   C("景阳冈", "第二十三回 横海郡柴进留宾 景阳冈武松打虎", "武松篇", "zigzag",
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```
13002        J(HeroId.WuSong), J(HeroId.ChaiJin, HeroId.SongJiang),
13003        "大虫", "景阳冈", BossSkill.Haste, HeroId.WuSong, "猎户",
13004        "武松在柴进庄上结识宋江，回乡探兄路过景阳冈。酒店挂着「三碗不过冈」，他偏喝了十八碗，提着哨棒上了冈·
13005        "武松打虎用的是双拳——哨棒一下打在枯树上折了，他索性揪住虎头一顿乱打。"),
13006    C("斗杀西门庆", "第二十六回 供人头武二郎设祭", "武松篇", "jiangzhou",
13007        J(HeroId.WuDaLang), J(HeroId.WuSong, HeroId.LeiHeng, HeroId.SongQing),
13008        "西门庆", "阳谷县财主", BossSkill.Resist, HeroId.WuSong, "王婆",
13009        "武松做了阳谷县都头，出差回来，哥哥武大郎已被潘金莲与西门庆毒死。他到狮子楼找西门庆，要替哥哥讨个公道
13010        "武大郎不在一百单八将之内，是本作的隐藏好汉：凑出他的名字，就有热乎的炊饼。"),
13011    C("十字坡", "第二十七回 母夜叉孟州道卖人肉 武都头十字坡遇张青", "武松篇", "qingfeng",
13012        J(HeroId.SunErNiang, HeroId.ZhangQingY), J(HeroId.WuSong),
13013        "孙二娘", "母夜叉", BossSkill.Haste, HeroId.WuSong, "公人",
13014        "武松刺配孟州，路过十字坡酒店。老板娘笑脸相迎，端上的酒却有股怪味——武松装作晕倒，看她怎么下手。",
13015        "张青告诫孙二娘：三等人不可坏——云游僧道、江湖上卖艺的、流配的犯人。"),
13016    C("醉打蒋门神", "第二十九回 施恩重霸孟州道 武松醉打蒋门神", "武松篇", "ring",
13017        J(HeroId.ShiEn), J(HeroId.WuSong, HeroId.ZhangQingY, HeroId.SunErNiang),
13018        "蒋门神", "快活林恶霸", BossSkill.Resist, HeroId.WuSong, "张团练",
13019        "施恩的快活林酒店被蒋门神强占。武松一路「无三不过望」，每过一家酒店喝三碗，醉醺醺地来到快活林。",
13020        "武松醉打蒋门神用的是「玉环步、鸳鸯脚」，是他平生得意的本事。"),
13021    C("血溅鸳鸯楼", "第三十一回 张都监血溅鸳鸯楼 武行者夜走蜈蚣岭", "武松篇", "jiuqu",
13022        J(HeroId.KongMing, HeroId.KongLiang), J(HeroId.WuSong, HeroId.ShiEn),
13023        "张都监", "孟州兵马都监", BossSkill.Summon, HeroId.WuSong, "张团练",
13024        "张都监设计陷害武松，又派人在飞云浦害他。武松杀回孟州城，血溅鸳鸯楼，扮作行者连夜出逃，在孔家庄醉打孔
13025        "血溅鸳鸯楼后，武松蘸血在白粉墙上写下八个大字：「杀人者，打虎武松也。」"),
13026    C("清风寨", "第三十三回 宋江夜看小鳌山 花荣大闹清风寨", "清风山篇", "river",
13027        J(HeroId.HuaRong, HeroId.YanShun, HeroId.WangYing, HeroId.ZhengTianShou), J(),
13028        "刘高", "清风寨文知寨", BossSkill.Summon, HeroId.HuaRong, "黄信",
13029        "宋江夜看花灯，被清风寨知寨刘高认出，五花大绑押进寨中。花荣连夜点兵：救哥哥！",
13030        "宋江在清风山差点被燕顺挖心做醒酒汤，一句「可惜宋江死在这里」才救了自己一命。"),
13031    C("瓦砾场", "第三十四回 镇三山大闹青州道 霹雳火夜走瓦砾场", "清风山篇", "plain",
13032        J(HeroId.QinMing, HeroId.HuangXin), J(HeroId.HuaRong, HeroId.YanShun, HeroId....
13033        "秦明", "霹雳火", BossSkill.Haste, HeroId.HuaRong, "慕容知府",
13034        "青州兵马总管霹雳火秦明奉命来剿清风山，一条狼牙棒杀得天昏地暗。花荣说：要擒此人，得用计。",
13035        "宋江派人扮作秦明去青州城下杀人放火，慕容知府杀了秦明全家，秦明从此死心塌地上了山。"),
13036    C("对影山", "第三十五回 石将军村店寄书 小李广梁山射雁", "清风山篇", "zigzag",
13037        J(HeroId.LvFang, HeroId.GuoSheng, HeroId.ShiYong), J(HeroId.HuaRong, HeroId.Q...
13038        "郭盛", "赛仁贵", BossSkill.Resist, HeroId.HuaRong, "吕方",
13039        "对影山上，小温侯吕方与赛仁贵郭盛争山头，画戟对画戟斗了几十天。两枝戟上的绒绦缠在了一起……",
13040        "花荣一箭射断缠在一起的绒绦，两人心服口服；上梁山后他又一箭射中雁行第三只。"),
13041    C("揭阳岭", "第三十六回 梁山泊吴用举戴宗 揭阳岭宋江逢李俊", "江州篇", "jiuqu",
13042        J(HeroId.LiLi, HeroId.LiJun, HeroId.TongWei, HeroId.TongMeng), J(HeroId.ShiYo...
13043        "李立", "催命判官", BossSkill.Summon, HeroId.LiJun, "穆春",
13044        "宋江刺配江州，在揭阳岭酒店喝了一碗酒就人事不省，被拖进作坊要开剥——混江龙李俊急匆匆赶到。",
13045        "揭阳三霸：岭上李俊、李立，镇上穆弘、穆春，江上张横、张顺。"),
13046    C("浔阳江", "第三十七回 没遮拦追赶及时雨 船火儿夜闹浔阳江", "江州篇", "bailong",
13047        J(HeroId.MuHong, HeroId.MuChun, HeroId.XueYong, HeroId.ZhangHeng), J(HeroId.L...
13048        "张横", "船火儿", BossSkill.Resist, HeroId.LiJun, "穆太公",
13049        "宋江赏了卖艺的薛永，得罪了揭阳镇的穆氏兄弟，被追到江边，慌忙跳上一条船——船家张横问：吃板刀面，还是
13050        "「板刀面」是一刀砍下水，「馄饨」是自己脱光跳下去。宋江报出名号，张横纳头便拜。"),
13051    C("浔阳楼", "第三十八回 及时雨会神行太保 黑旋风斗浪里白条", "江州篇", "marsh",
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13052       J(HeroId.DaiZong, HeroId.LiKui, HeroId.ZhangShun), J(HeroId.ZhangHeng, HeroId...
13053       "张顺", "浪里白条", BossSkill.Haste, HeroId.ZhangHeng, "戴宗",
13054       "李逵在江边抢鱼，和渔牙子张顺打了起来。岸上张顺打不过，一把把李逵拖下水——这下轮到铁牛喝水了。",
13055       "黑旋风斗浪里白条，是书中最有名的「不打不相识」：两人后来成了生死兄弟。"),
13056   C("江州法场", "第四十回 梁山泊好汉劫法场 白龙庙英雄小聚义", "江州篇", "jiangzhou",
13057       J(HeroId.SongJiang, HeroId.XiaoRang, HeroId.JinDaJian), J(HeroId.LiKui, HeroI...
13058       "蔡九", "江州知府", BossSkill.Haste, HeroId.LiKui, "晁盖",
13059       "浔阳楼上一首反诗，宋江被判斩首。萧让、金大坚仿造的蔡京书信露了破绽。午时三刻，十字路口，刽子手已经举
13060       "宋江的反诗：「他时若遂凌云志，敢笑黄巢不丈夫。」黄巢是唐末造反的首领。"),
13061   C("白龙庙", "第四十一回 宋江智取无为军 张顺活捉黄文炳", "江州篇", "bailong",
13062       J(HeroId.HouJian), J(HeroId.ZhangShun, HeroId.ZhangHeng, HeroId.RuanXiaoEr, H...
13063       "黄文炳", "无为军通判", BossSkill.Resist, HeroId.ZhangShun, "李俊",
13064       "劫了法场，众好汉护着宋江退到江边白龙庙。告发反诗的黄文炳就在对岸无为军，裁缝侯健说：他家的路，我熟。
13065       "白龙庙小聚义之后，宋江正式上了梁山，坐了第二把交椅。"),
13066   C("黄门山", "第四十一回 回山途中", "江州篇", "zigzag",
13067       J(HeroId.OuPeng, HeroId.JiangJing, HeroId.MaLin, HeroId.TaoZongWang), J(HeroI...
13068       "欧鹏", "摩云金翅", BossSkill.Haste, HeroId.HuaRong, "蒋敬",
13069       "回梁山的路上，黄门山四个头领拦住去路。一问来的是及时雨宋公明，四人当即下山相迎。",
13070       "蒋敬是梁山的「账房」，书中说他「精通书算，积万累千，纤毫不差」。"),
13071   C("沂岭杀四虎", "第四十三回 假李逵剪径劫单人 黑旋风沂岭杀四虎", "李逵篇", "ring",
13072       J(HeroId.ZhuFu, HeroId.LiYun), J(HeroId.LiKui, HeroId.ZhuGui),
13073       "大虫", "沂岭", BossSkill.Haste, HeroId.LiKui, "李鬼",
13074       "李逵回家接老娘上山享福，背到沂岭，老娘口渴，他去打水回来，只见一摊血……",
13075       "李逵怒杀四虎后被捉，朱富用蒙汗药酒麻翻了自己的师父李云，救出李逵，师徒同上梁山。"),
13076   C("饮马川", "第四十四回 锦豹子小径逢戴宗 病关索长街遇石秀", "祝家庄篇", "snake",
13077       J(HeroId.YangLin, HeroId.DengFei, HeroId.MengKang, HeroId.PeiXuan), J(HeroId....
13078       "邓飞", "火眼狻猊", BossSkill.Haste, HeroId.YangLin, "裴宣",
13079       "戴宗下山打探公孙胜，路遇锦豹子杨林，一同走到饮马川，山上冲下一伙强人——为首的火眼狻猊邓飞，正是杨林
13080       "饮马川原是邓飞、孟康的山寨，铁面孔目裴宣来了之后被推为寨主。"),
13081   C("翠屏山", "第四十六回 病关索大闹翠屏山 拼命三火烧祝家店", "祝家庄篇", "river",
13082       J(HeroId.YangXiong, HeroId.ShiXiu, HeroId.ShiQian), J(HeroId.YangLin),
13083       "裴如海", "报恩寺僧", BossSkill.Resist, HeroId.ShiXiu, "潘巧云",
13084       "拼命三郎石秀看破了嫂嫂与和尚裴如海私通，杨雄翠屏山上清理门户，带着时迁投奔梁山，路上住进了祝家店……
13085       "时迁在祝家店偷了报晓的公鸡下酒，引出了三打祝家庄。"),
13086   C("一打祝家庄", "第四十七回 扑天雕双修生死书 宋公明一打祝家庄", "祝家庄篇", "jiuqu",
13087       J(HeroId.LiYing, HeroId.DuXing), J(HeroId.YangXiong, HeroId.ShiXiu, HeroId.Sh...
13088       "祝龙", "祝家庄大郎", BossSkill.Summon, HeroId.LiYing, "祝朝奉",
13089       "时迁被祝家庄捉了。李家庄李应写信求情，反被祝彪一箭射伤。宋江起兵来打，却在盘陀路里转得晕头转向。",
13090       "祝家庄的盘陀路，要见白杨树才转弯，否则怎么走都走不出去——梁山第一次就吃了这个亏。"),
13091   C("二打祝家庄", "第四十八回 一丈青单捉王矮虎 宋公明两打祝家庄", "祝家庄篇", "ring",
13092       J(HeroId.HuSanNiang), J(HeroId.LinChong, HeroId.WangYing, HeroId.LiYing),
13093       "扈三娘", "扈家庄", BossSkill.Haste, HeroId.LinChong, "扈成",
13094       "扈家庄的一丈青扈三娘使日月双刀，一阵就活捉了矮脚虎王英。林冲拍马迎上……",
13095       "梁山三位女将：一丈青扈三娘、母夜叉孙二娘、母大虫顾大嫂。"),
13096   C("登州劫牢", "第四十九回 解珍解宝双越狱 孙立孙新大劫牢", "祝家庄篇", "plain",
13097       J(HeroId.XieZhen, HeroId.XieBao, HeroId.GuDaSao, HeroId.SunXin), J(),
13098       "毛太公", "登州恶霸", BossSkill.Summon, HeroId.XieZhen, "毛仲义",
13099       "猎户解珍、解宝射死的老虎滚进了毛太公家后园，毛太公不但不还，还诬他们是贼关进死牢。表姐顾大嫂拍案：
13100       "顾大嫂一句「你们不去，我便自去」，逼得做提辖的孙立也一起劫了登州大牢。"),
13101   C("三打祝家庄", "第五十回 吴学究双掌连环计 宋公明三打祝家庄", "祝家庄篇", "jiuqu",
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        J(HeroId.SunLi, HeroId.YueHe, HeroId.ZouYuan, HeroId.ZouRun), J(HeroId.XieBao),
        "栾廷玉", "祝家庄教师", BossSkill.Resist, HeroId.SunLi, "祝彪",
        "孙立是祝家庄教师栾廷玉的同门师兄弟。他打着官府的旗号「来助战」，开进了祝家庄……",
        "孙立里应外合，三打祝家庄终于成功。这是梁山第一次打下有城墙的庄子。"),
    C("高唐州", "第五十四回 入云龙斗法破高廉 黑旋风探穴救柴进", "高唐州篇", "zigzag",
        J(HeroId.TangLong), J(HeroId.GongSunSheng, HeroId.LiKui, HeroId.ChaiJin),
        "高廉", "高唐州知府", BossSkill.Summon, HeroId.GongSunSheng, "殷天锡",
        "柴进被高俅的堂弟高廉关进死牢。高廉会妖法，梁山连败两阵。李逵、戴宗去蓟州请回公孙胜，路上又结识了铁
        "公孙胜的师父罗真人曾把李逵丢到半空，摔进蓟州府牢里，吓得铁牛再不敢胡来。"),
    C("连环马", "第五十七回 徐宁教使钩镰枪 宋江大破连环马", "连环马篇", "plain",
        J(HeroId.XuNing, HeroId.HanTao, HeroId.PengQi, HeroId.LingZhen), J(HeroId.Tan...
        "呼延灼", "双鞭", BossSkill.Haste, HeroId.XuNing, "高俅",
        "朝廷派呼延灼带连环马来剿梁山，三十匹马用铁环连成一排，冲到哪里哪里倒。汤隆说：只有我表哥徐宁的钩镰
        "时迁连夜盗走徐宁家传的雁翎锁子甲，把他赚上梁山。钩镰枪专钩马腿，连环马便破了。"),
    C("三山聚义", "第五十八回 三山聚义打青州 众虎同心归水泊", "连环马篇", "ring",
        J(HeroId.HuYanZhuo), J(HeroId.LuZhiShen, HeroId.YangZhi, HeroId.WuSong, HeroI...
        "慕容知府", "青州知府", BossSkill.Resist, HeroId.QinMing, "桃花山",
        "呼延灼兵败投奔青州，替慕容知府去打二龙山、桃花山、白虎山。三山好汉合兵一处，请宋江来破青州。",
        "三山聚义：二龙山、桃花山、白虎山同归梁山，梁山的实力从此大增。"),
    C("芒砀山", "第六十回 公孙胜芒砀山降魔", "连环马篇", "ring",
        J(HeroId.FanRui, HeroId.XiangChong, HeroId.LiGun), J(HeroId.GongSunSheng),
        "樊瑞", "混世魔王", BossSkill.Summon, HeroId.GongSunSheng, "项充",
        "芒砀山混世魔王樊瑞扬言要吞并梁山。他会呼风唤雨，手下项充、李衮团牌飞刀密不透风。",
        "樊瑞被公孙胜破了妖法，心悦诚服，拜公孙胜为师。"),
    C("曾头市", "第六十回 晁天王曾头市中箭", "曾头市篇", "jiangzhou",
        J(HeroId.DuanJingZhu), J(HeroId.LinChong, HeroId.HuYanZhuo, HeroId.HuaRong),
        "史文恭", "曾头市教师", BossSkill.Resist, HeroId.HuaRong, "曾长官",
        "段景住盗来的照夜玉狮子马被曾头市夺去，曾家还编歌辱骂梁山。晁盖大怒，亲自领兵去打曾头市……",
        "晁盖中了史文恭的毒箭，临终遗言：「若哪个捉得射死我的，便叫他做梁山泊主。」"),
    C("智赚玉麒麟", "第六十一回 吴用智赚玉麒麟 张顺夜闹金沙渡", "大名府篇", "snake",
        J(HeroId.LuJunYi, HeroId.YanQing), J(HeroId.WuYong, HeroId.LiKui, HeroId.Zhan...
        "李固", "卢府管家", BossSkill.Haste, HeroId.YanQing, "贾氏",
        "吴用扮作算命先生，李逵扮成哑道童，到大名府给卢俊义算了一卦：百日之内有血光之灾，须往东南千里外躲避·
        "吴用让卢俊义在墙上题的四句诗：「芦花丛里一扁舟，俊杰俄从此地游。义士若能知此理，反躬逃难可无忧。」藏
    C("石秀跳楼", "第六十二回 放冷箭燕青救主 劫法场石秀跳楼", "大名府篇", "qingfeng",
        J(HeroId.CaiFu, HeroId.CaiQing), J(HeroId.ShiXiu, HeroId.LuJunYi, HeroId.YanQ...
        "梁中书", "大名府留守", BossSkill.Summon, HeroId.ShiXiu, "李固",
        "卢俊义被李固陷害，要在大名府问斩。刽子手蔡福、蔡庆受了柴进的银子，刀下留情；楼上，一个人纵身跳了下来
        "石秀独自跳楼劫法场，大喊「梁山泊好汉全伙在此」，吓得官兵四散。"),
    C("关胜议取梁山", "第六十三回 宋江兵打北京城 关胜议取梁山泊", "大名府篇", "snake",
        J(HeroId.GuanSheng, HeroId.XuanZan, HeroId.HaoSiWen), J(HeroId.HuYanZhuo, Her...
        "关胜", "大刀", BossSkill.Resist, HeroId.HuYanZhuo, "蔡京",
        "梁山兵围大名府。宣赞举荐关云长的后人关胜，关胜献计围魏救赵，直扑梁山老巢。",
        "呼延灼诈降，月夜把关胜赚进了梁山的埋伏，关胜感宋江义气，归顺梁山。"),
    C("雪天擒索超", "第六十四回 呼延灼月夜赚关胜 宋公明雪天擒索超", "大名府篇", "zigzag",
        J(HeroId.SuoChao), J(HeroId.YangZhi, HeroId.GuanSheng),
        "索超", "急先锋", BossSkill.Haste, HeroId.YangZhi, "闻达",
        "大雪纷飞，大名府急先锋索超又冲出城来。吴用笑道：此人性急，雪地里挖个陷坑就够了。",
        "索超当年在大名府校场与杨志大战五十回合不分胜负，梁中书把两人都升了官。"),
    C("张顺请神医", "第六十五回 托塔天王梦中显圣 浪里白条水上报冤", "大名府篇", "marsh",
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13152       J(HeroId.AnDaoQuan, HeroId.WangDingLiu), J(HeroId.ZhangShun),
13153       "张旺", "截江鬼", BossSkill.Summon, HeroId.ZhangShun, "孙五",
13154       "宋江背上生了恶疮。张顺星夜过江去建康请神医安道全，船到江心，梢公张旺抽出了刀……",
13155       "张顺被绑住丢下江，在水底咬断绳索逃生；王定六收留了他，后来一同报了仇。"),
13156   C("水火二将", "第六十七回 宋江赏马步三军 关胜降水火二将", "大名府篇", "river",
13157       J(HeroId.ShanTingGui, HeroId.WeiDingGuo, HeroId.BaoXu, HeroId.JiaoTing), J(He...
13158       "魏定国", "神火将", BossSkill.Summon, HeroId.GuanSheng, "单廷珪",
13159       "凌州的圣水将、神火将来攻梁山。关胜领兵迎敌；李逵偷偷下山，在枯树山收了鲍旭、焦挺。",
13160       "单廷珪会用「水浸兵」，魏定国会用「火攻兵」，都被关胜一一说降。"),
13161   C("夜打曾头市", "第六十八回 宋公明夜打曾头市 卢俊义活捉史文恭", "曾头市篇", "plain",
13162       J(HeroId.YuBaoSi), J(HeroId.LuJunYi, HeroId.YanQing, HeroId.ShiQian, HeroId.D...
13163       "史文恭", "曾头市教师", BossSkill.Resist, HeroId.LuJunYi, "曾涂",
13164       "险道神郁保四又劫了段景住的马送到曾头市。宋江再打曾头市，为晁盖报仇。",
13165       "卢俊义活捉史文恭为晁盖报了仇，按遗言本当做寨主，他却坚决推让给宋江。"),
13166   C("东平府", "第六十九回 东平府误陷九纹龙 宋公明义释双枪将", "东平东昌篇", "qingfeng",
13167       J(HeroId.DongPing), J(HeroId.ShiJin, HeroId.HuSanNiang, HeroId.WangYing, Hero...
13168       "董平", "双枪将", BossSkill.Haste, HeroId.HuSanNiang, "程太守",
13169       "宋江、卢俊义拈阄分兵，谁先打下城池谁做寨主。宋江打东平府，史进潜入城中却被识破，双枪将董平杀出城来·
13170       "董平的旗上写着：「英雄双枪将，风流万户侯。」"),
13171   C("东昌府", "第七十回 没羽箭飞石打英雄 宋公明弃粮擒壮士", "东平东昌篇", "river",
13172       J(HeroId.ZhangQing, HeroId.GongWang, HeroId.DingDeSun, HeroId.HuangFuDuan), J...
13173       "张清", "没羽箭", BossSkill.Resist, HeroId.RuanXiaoQi, "龚旺",
13174       "东昌府守将没羽箭张清，一颗颗石子飞出，连伤梁山十五员大将。吴用说：岸上打不过，就引他到水边。",
13175       "张清被擒后，宋江惜才不杀，又让他举荐了兽医皇甫端。至此一百单八将全部聚齐。"),
13176   C("智扑擎天柱", "第七十四回 燕青智扑擎天柱 李逵寿张乔坐衙", "排座次篇", "jiuqu",
13177       J(), J(HeroId.YanQing, HeroId.LiKui, HeroId.LuJunYi, HeroId.SongJiang),
13178       "任原", "擎天柱", BossSkill.Resist, HeroId.YanQing, "部署",
13179       "忠义堂上排定座次之后，浪子燕青独自下山，要在泰安州的相扑擂台上，摔下两年无敌的擎天柱任原。",
13180       "忠义堂前天降石碣，上刻三十六天罡、七十二地煞的星号与姓名，这就是「梁山泊英雄排座次」。"),
13181   C("三败高俅", "第八十回 张顺凿漏海鳅船 宋江三败高太尉", "招安篇", "marsh",
13182       J(), J(HeroId.ZhangShun, HeroId.RuanXiaoEr, HeroId.RuanXiaoWu, HeroId.RuanXia...
13183       "高俅", "殿帅太尉", BossSkill.Summon, HeroId.ZhangShun, "童贯",
13184       "高俅亲率大军和海鳅大船来攻梁山。张顺带水军潜到船底，一凿一凿……",
13185       "高俅落水被擒，林冲恨不得生吃了他，宋江却亲自松绑——一心只想招安。"),
13186   C("征辽", "第八十三回 宋公明奉诏破大辽", "征辽篇", "plain",
13187       J(), J(HeroId.GuanSheng, HeroId.LinChong, HeroId.QinMing, HeroId.HuYanZhuo, H...
13188       "兀颜光", "辽国统军", BossSkill.Resist, HeroId.GuanSheng, "辽主",
13189       "梁山全伙受了招安，改称宋军。第一仗就是北上征辽，辽国统军兀颜光摆下了太乙混天象阵。",
13190       "五虎上将关胜、林冲、秦明、呼延灼、董平，征辽时各领一军，杀得辽兵望风而逃。"),
13191   C("征田虎", "第九十一回起 征田虎", "征田虎篇", "zigzag",
13192       J(), J(HeroId.GongSunSheng, HeroId.FanRui, HeroId.LuJunYi, HeroId.YanQing),
13193       "乔道清", "幻魔君", BossSkill.Summon, HeroId.GongSunSheng, "田虎",
13194       "河北田虎造反，他手下的幻魔君乔道清会妖法，宋军接连吃亏。公孙胜仗剑出阵。",
13195       "征田虎、征王庆是一百二十回本才有的情节；这两路征战中，一百单八将无一阵亡。"),
13196   C("征王庆", "第一百零一回起 征王庆", "征王庆篇", "bailong",
13197       J(), J(HeroId.LiJun, HeroId.ZhangHeng, HeroId.ZhangShun, HeroId.TongWei, Hero...
13198       "王庆", "淮西反王", BossSkill.Haste, HeroId.LiJun, "段三娘",
13199       "淮西王庆兵败，乘船逃命。江面上，混江龙李俊早已等候多时。",
13200       "王庆是被李俊在江上擒获的。梁山水军从此威名远扬。"),
13201   C("涌金门", "第一百一十四回 宁海军宋江吊孝 涌金门张顺归神", "征方腊篇", "marsh",
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            J(), J(HeroId.ZhangShun, HeroId.ZhangHeng, HeroId.RuanXiaoQi, HeroId.LiJun),
            "方天定", "方腊太子", BossSkill.Resist, HeroId.ZhangHeng, "石宝",
            "征方腊打到杭州，城高难攻。张顺说：我从西湖水底潜到涌金门，放火为号——",
            "张顺在涌金门被乱箭射死，魂魄附在哥哥张横身上，亲手杀了方天定。"),
        C("乌龙岭", "第一百一十七回 睦州城箭射邓元觉 乌龙岭神助宋公明", "征方腊篇", "jiuqu",
            J(), J(HeroId.HuaRong, HeroId.XieZhen, HeroId.XieBao, HeroId.SongJiang, HeroI...
            "邓元觉", "宝光国师", BossSkill.Resist, HeroId.HuaRong, "方腊",
            "乌龙岭山高路险，解珍、解宝扮作猎户攀崖探路，再没回来。睦州城下，宝光国师邓元觉挥着禅杖杀出……",
            "征方腊一路，梁山好汉死伤大半，是全书最悲壮的一段。"),
        C("擒方腊", "第一百一十八回 卢俊义大战昱岭关 宋公明智取清溪洞", "征方腊篇", "ring",
            J(), J(HeroId.LuZhiShen, HeroId.WuSong, HeroId.LuJunYi, HeroId.SongJiang, Her...
            "方腊", "圣公", BossSkill.Summon, HeroId.LuZhiShen, "方杰",
            "清溪洞破，方腊只身逃进深山，迎面撞上一个胖大和尚，一禅杖打翻在地。",
            "鲁智深生擒方腊，应了智真长老当年的偈语「逢夏而擒，遇腊而执」。后来他在六和寺听潮信圆寂。"),
        C("蓼儿洼", "第一百二十回 宋公明神聚蓼儿洼 徽宗帝梦游梁山泊", "终章", "marsh",
            J(), J(HeroId.SongJiang, HeroId.LiKui, HeroId.WuYong, HeroId.HuaRong, HeroId....
            "蔡京", "太师", BossSkill.Summon, HeroId.LiKui, "高俅",
            "征方腊归来，奸臣蔡京、高俅、童贯、杨戬容不下梁山好汉，一杯御赐药酒送到了楚州。替天行道的大旗，最后一
            "宋江饮了毒酒，怕李逵再反坏了梁山的忠义之名，叫来李逵同饮，一起葬在楚州蓼儿洼。吴用、花荣也在墓前自缢
    };
    int n = list.Length;
    for (int i = 0; i < n; i++)
    {
        list[i].Index = i;
        float k = Math.Min(1f, i / 45f);
        list[i].RivalSkill = 0.25f + 0.65f * k;
        list[i].RivalIncome = 0.6f + 0.4f * k;
        list[i].RivalFoeHp = 1f;
        list[i].FoeHp = 1f + 0.35f * k;
    }
    return list;
}
/// <summary>某好汉在哪一回入伙；-1 表示剧情里不入伙。</summary>
public static int JoinChapter(HeroId id)
{
    foreach (var ch in All)
        if (Array.IndexOf(ch.Joins, id) >= 0) return ch.Index;
    return -1;
}
/// <summary>第一次出场的回。</summary>
public static int FirstSeen(HeroId id)
{
    foreach (var ch in All)
        if (Array.IndexOf(ch.Pool, id) >= 0) return ch.Index;
    return -1;
}
// ———————————————————————————————— 聚义厅 ————————————————————————————————
public const int TeamSize = 5;
/// <summary>
/// 剧情打完后的车轮战：阵容由玩家自选（或随机）五员，名字里的字只从这五人里出，
```

```
13252       /// 同屏金字不至于太杂；对手和头目每阵变强。
13253       /// </summary>
13254       public static ChapterDef Endless(int round, IList<HeroId> team, uint seed)
13255       {
13256           var rng = new Rng(seed * 7919u + (uint)round * 104729u + 1u);
13257           var pool = new List<HeroId>();
13258           foreach (var id in team) if (!pool.Contains(id) && !HeroDef.Get(id).Hidden) pool.Add(...
13259           if (pool.Count == 0) pool.Add(HeroId.SongJiang);
13260           var fighters = new List<HeroId>();
13261           foreach (var id in pool) if (!HeroDef.Get(id).NoAttack) fighters.Add(id);
13262           if (fighters.Count == 0) { pool.Add(HeroId.LinChong); fighters.Add(HeroId.LinChong); }
13263           var src = All[rng.Range(0, All.Length)].Boss;
13264           var nemesis = fighters[rng.Range(0, fighters.Count)];
13265           var skill = (BossSkill)rng.Range(0, 3);
13266           var rivalPick = HeroDef.All[rng.Range(16, HeroDef.All.Length)];
13267           float k = Math.Min(1f, (round - 1) / 20f);
13268           return new ChapterDef
13269           {
13270               Index = All.Length,
13271               Round = round,
13272               Name = "聚义厅 · 第" + round + "阵",
13273               Hui = "替天行道",
13274               Act = "聚义厅",
13275               Story = $"一百单八将聚齐，各路英雄来梁山讨教。这一阵来的是{src.Title}{src.Name}，看你点的五员大将能撑几阵!
13276               Lore = "每阵结束都可以换将。阵容里有渊源的好汉同时上场，会触发「羁绊」，伤害更高。",
13277               Map = Maps[EndlessMaps[rng.Range(0, EndlessMaps.Length)]],
13278               Pool = pool.ToArray(),
13279               Boss = new BossDef { Name = src.Name, Title = src.Title, Skill = skill, Nemesis =...
13280               RivalName = rivalPick.Nick + rivalPick.Name,
13281               RivalSkill = 0.85f + 0.15f * k,
13282               RivalIncome = 0.95f + 0.15f * k,
13283               FoeHp = 1.25f + 0.3f * k,
13284           };
13285       }
13286       /// <summary>同一回换一池好汉（地图、头目、对手不变）。</summary>
13287       public ChapterDef WithPool(params HeroId[] pool)
13288       {
13289           var c = (ChapterDef)MemberwiseClone();
13290           c.Pool = pool;
13291           return c;
13292       }
13293   }
13294 }
13295 // ===== FILE: Assets/Scripts/Sim/Events.cs =====
13296 namespace Rescue.Sim
13297 {
13298   public enum Ev
13299   {
13300       Recruit, Merge, HeroFormed, Attack, HeroAttack, Hit, Kill, Leak, Rain, Ult,
13301       WaveStart, BossSpawn, EliteWarn, Dig, Buns, ItemUsed, End, Dismiss,
```

```
13302       HeroLevel, Shovels, Rescue, Baozi, Bond, Heal, Steal,
13303   }
13304   /// <summary>A 通常是棋子 / 敌人 / 好汉的 Id；P、Q 是所在半区自己的格子坐标。</summary>
13305   public struct SimEvent
13306   {
13307       public Ev Type;
13308       public int Side;
13309       public int A, B;
13310       public V2 P, Q;
13311       public float Val;
13312       public int Flag;
13313       public string Text;
13314   }
13315 }
13316 // ===== FILE: Assets/Scripts/Sim/RivalAI.cs =====
13317 using System;
13318 using System.Collections.Generic;
13319 namespace Rescue.Sim
13320 {
13321   /// <summary>对手：只通过和玩家相同的操作接口下棋。skill 0~1 决定思考间隔和失误率。</summary>
13322   public sealed class RivalAI
13323   {
13324       readonly Side _s;
13325       readonly float _skill;
13326       readonly Rng _rng;
13327       float _think = 1.5f;
13328       readonly float[][,] _coverage = new float[4][,];
13329       public RivalAI(Side side, float skill, uint seed)
13330       {
13331           _s = side;
13332           _skill = Math.Max(0, Math.Min(1, skill));
13333           _rng = new Rng(seed);
13334           for (int k = 0; k < 4; k++) _coverage[k] = BuildCoverage(UnitDef.Get((UnitKind)k).Ran...
13335       }
13336       float[,] BuildCoverage(float range)
13337       {
13338           var b = _s.Board;
13339           var map = new float[Board.Cols, Board.Rows];
13340           for (int c = 0; c < Board.Cols; c++)
13341               for (int r = 0; r < Board.Rows; r++)
13342               {
13343                   var p = new CellPos(c, r).Center;
13344                   float s = 0;
13345                   for (int i = 0; i < b.Waypoints.Length; i++)
13346                       if (V2.Dist(p, b.Waypoints[i]) <= range) s += 1f + 0.6f * i / b.Waypoints...
13347                   map[c, r] = s;
13348               }
13349           return map;
13350       }
13351       public void Tick(float dt)
```

```
13352       {
13353           if (_s.Dead || _s.M.Over) return;
13354           _think -= dt;
13355           if (_think > 0) return;
13356           float baseT = 1.7f - 1.25f * _skill;
13357           _think = baseT * (0.75f + 0.5f * _rng.Next01());
13358           Act();
13359       }
13360       void Act()
13361       {
13362           if (TryDig()) return;
13363           if (_rng.Chance(0.15f + 0.85f * _skill) && TryMergeCamp()) return;
13364           for (int i = 0; i < _s.Camp.Length; i++)
13365           {
13366               var p = _s.Camp[i];
13367               if (p == null) continue;
13368               if (p.Type == PieceType.Unit && TryPlaceUnit(i, p)) return;
13369               if (p.Type == PieceType.HeroChar && TryPlaceChar(i, p)) return;
13370           }
13371           // 一次聚义填满所有空位且价格递增：高手等山寨快空了再按，新手见空就按
13372           int empty = _s.EmptyCampSlots;
13373           int wantEmpty = _skill > 0.5f ? 4 : 2;
13374           if (_s.CanRecruit && empty >= wantEmpty) { _s.Recruit(); return; }
13375           if (_rng.Chance(0.3f + 0.7f * _skill) && TryMergeBoard()) return;
13376           if (_rng.Chance(_skill) && TryUpgradeSwap()) return;
13377           if (_rng.Chance(_skill) && TryClearOrphan()) return;
13378           if (_s.CanRecruit && empty > 0 && (_s.Buns >= _s.RecruitCost * 2 || _rng.Chance(0.5f ...
13379           if (empty == 0 && _rng.Chance(0.2f + 0.8f * _skill)) TryDismiss();
13380       }
13381       /// <summary>山寨堵死时遣散最没用的字（高手挑，新手随手扔）。</summary>
13382       void TryDismiss()
13383       {
13384           int pick = -1;
13385           float worst = float.MaxValue;
13386           for (int i = 0; i < _s.Camp.Length; i++)
13387           {
13388               var p = _s.Camp[i];
13389               if (p == null) continue;
13390               float v = p.Type == PieceType.Shovel ? 100 : p.Type == PieceType.HeroChar ? (Best...
13391               v += _rng.Next01() * (1f - _skill) * 40f;
13392               if (v < worst) { worst = v; pick = i; }
13393           }
13394           if (pick >= 0) _s.DismissCamp(pick);
13395       }
13396       /// <summary>有铲子就往离路最近的草地开荒；新手挑得随便。</summary>
13397       bool TryDig()
13398       {
13399           var b = _s.Board;
13400           for (int i = 0; i < _s.Camp.Length; i++)
13401           {
```

```
13402            var p = _s.Camp[i];
13403            if (p == null || p.Type != PieceType.Shovel) continue;
13404            int bc = -1, br = -1;
13405            float best = -1;
13406            for (int c = 0; c < Board.Cols; c++)
13407               for (int r = 0; r < Board.Rows; r++)
13408               {
13409                  if (!b.Diggable(c, r)) continue;
13410                  float s = _coverage[0][c, r] + _coverage[2][c, r] * 0.5f + _rng.Next01() ...
13411                  if (s > best) { best = s; bc = c; br = r; }
13412               }
13413            if (bc >= 0) return _s.CampToCell(i, bc, br);
13414            _s.DismissCamp(i);
13415            return true;
13416         }
13417         return false;
13418      }
13419      bool TryPlaceUnit(int slot, Piece p)
13420      {
13421         var b = _s.Board;
13422         if (_rng.Chance(0.25f + 0.75f * _skill))
13423            for (int c = 0; c < Board.Cols; c++)
13424               for (int r = 0; r < Board.Rows; r++)
13425                  if (Side.CanMerge(p, b.Pieces[c, r])) return _s.CampToCell(slot, c, r);
13426         var cov = _coverage[(int)p.Kind];
13427         int bc = -1, br = -1;
13428         float best = -1;
13429         for (int c = 0; c < Board.Cols; c++)
13430            for (int r = 0; r < Board.Rows; r++)
13431            {
13432               if (!b.Placeable(c, r) || b.Pieces[c, r] != null) continue;
13433               float s = cov[c, r] + _rng.Next01() * (1f - _skill) * 12f;
13434               if (s > best) { best = s; bc = c; br = r; }
13435            }
13436         return bc >= 0 && _s.CampToCell(slot, bc, br);
13437      }
13438      bool TryPlaceChar(int slot, Piece p)
13439      {
13440         var b = _s.Board;
13441         for (int c = 0; c < Board.Cols; c++)
13442            for (int r = 0; r < Board.Rows; r++)
13443               if (Side.CanMerge(p, b.Pieces[c, r]) && _s.HeroAt(c, r) != null) return _s.Ca...
13444         bool smart = _rng.Chance(0.35f + 0.65f * _skill);
13445         if (smart)
13446         {
13447            NameMatcher.ReadDirs(_s.Mirrored, out var dh, out var dv);
13448            int bestScore = -1;
13449            CellPos bestCell = default;
13450            foreach (var def in _s.Pool)
13451            {
```

```
            if (def.Name.IndexOf(p.Char) < 0) continue;
            foreach (var d in new[] { dh, dv })
               for (int c = 0; c < Board.Cols; c++)
                  for (int r = 0; r < Board.Rows; r++)
                  {
                     int score = ScorePlacement(def.Name, p.Char, c, r, d, out var at);
                     if (score > bestScore) { bestScore = score; bestCell = at; }
                  }
         }
         if (bestScore >= 0) return _s.CampToCell(slot, bestCell.c, bestCell.r);
      }
      if (_s.EmptyCampSlots > 0 || _rng.Chance(_skill)) return false;
      for (int c = 0; c < Board.Cols; c++)
         for (int r = 0; r < Board.Rows; r++)
            if (b.Placeable(c, r) && b.Pieces[c, r] == null) return _s.CampToCell(slot, c...
      return false;
   }
   int ScorePlacement(string name, char ch, int c, int r, CellPos d, out CellPos at)
   {
      at = default;
      var b = _s.Board;
      int have = 0;
      bool slotFound = false;
      for (int i = 0; i < name.Length; i++)
      {
         int cc = c + d.c * i, rr = r + d.r * i;
         if (!b.Placeable(cc, rr)) return -1;
         var q = b.Pieces[cc, rr];
         if (q == null)
         {
            if (name[i] == ch && !slotFound) { slotFound = true; at = new CellPos(cc, rr); }
            continue;
         }
         if (q.Type == PieceType.HeroChar && q.Char == name[i] && _s.HeroAt(cc, rr) == nul...
         return -1;
      }
      if (!slotFound) return -1;
      return have * 10 + (int)_coverage[0][at.c, at.r];
   }
   int BestCharScore(char ch)
   {
      NameMatcher.ReadDirs(_s.Mirrored, out var dh, out var dv);
      int best = -1;
      foreach (var def in _s.Pool)
      {
         if (def.Name.IndexOf(ch) < 0) continue;
         foreach (var d in new[] { dh, dv })
            for (int c = 0; c < Board.Cols; c++)
               for (int r = 0; r < Board.Rows; r++)
                  best = Math.Max(best, ScorePlacement(def.Name, ch, c, r, d, out _));
```

```
13502         }
13503         return best;
13504     }
13505     /// <summary>棋盘满了：清掉场上凑不成名字的孤字，给兵腾地方。</summary>
13506     bool TryClearOrphan()
13507     {
13508         var b = _s.Board;
13509         bool campHasUnit = false;
13510         foreach (var p in _s.Camp) if (p != null && p.Type == PieceType.Unit) campHasUnit = t...
13511         if (!campHasUnit) return false;
13512         for (int c = 0; c < Board.Cols; c++)
13513             for (int r = 0; r < Board.Rows; r++)
13514                 if (b.Placeable(c, r) && b.Pieces[c, r] == null) return false;
13515         for (int c = 0; c < Board.Cols; c++)
13516             for (int r = 0; r < Board.Rows; r++)
13517             {
13518                 var q = b.Pieces[c, r];
13519                 if (q == null || q.Type != PieceType.HeroChar || _s.HeroAt(c, r) != null) con...
13520                 bool partner = false;
13521                 foreach (var n in b.Neighbors4(new CellPos(c, r)))
13522                 {
13523                     var o = b.Pieces[n.c, n.r];
13524                     if (o == null || o.Type != PieceType.HeroChar) continue;
13525                     foreach (var def in _s.Pool)
13526                         if (def.Name.IndexOf(q.Char) >= 0 && def.Name.IndexOf(o.Char) >= 0) p...
13527                 }
13528                 if (!partner) return _s.DismissCell(c, r);
13529             }
13530         return false;
13531     }
13532     bool TryMergeCamp()
13533     {
13534         var camp = _s.Camp;
13535         for (int i = 0; i < camp.Length; i++)
13536             for (int j = i + 1; j < camp.Length; j++)
13537                 if (camp[i] != null && camp[i].Type == PieceType.Unit && Side.CanMerge(camp[i...
13538                     return _s.CampToCamp(i, j);
13539         return false;
13540     }
13541     /// <summary>棋盘满了：用山寨里更高级的兵换下场上同类里最弱的那个。</summary>
13542     bool TryUpgradeSwap()
13543     {
13544         var b = _s.Board;
13545         for (int c = 0; c < Board.Cols; c++)
13546             for (int r = 0; r < Board.Rows; r++)
13547                 if (b.Placeable(c, r) && b.Pieces[c, r] == null) return false;
13548         for (int i = 0; i < _s.Camp.Length; i++)
13549         {
13550             var p = _s.Camp[i];
13551             if (p == null || p.Type != PieceType.Unit) continue;
```

```
            int bc = -1, br = -1, worst = p.Level;
            for (int c = 0; c < Board.Cols; c++)
               for (int r = 0; r < Board.Rows; r++)
               {
                  var q = b.Pieces[c, r];
                  if (q == null || q.Type != PieceType.Unit || q.Level >= worst) continue;
                  worst = q.Level; bc = c; br = r;
               }
            if (bc >= 0) return _s.CampToCell(i, bc, br);
         }
         return false;
      }
      bool TryMergeBoard()
      {
         var b = _s.Board;
         var list = new List<CellPos>();
         for (int c = 0; c < Board.Cols; c++)
            for (int r = 0; r < Board.Rows; r++)
               if (b.Pieces[c, r] != null && b.Pieces[c, r].Type == PieceType.Unit) list.Add...
         list.Sort((x, y) => b.Pieces[x.c, x.r].Level.CompareTo(b.Pieces[y.c, y.r].Level));
         for (int i = 0; i < list.Count; i++)
            for (int j = 0; j < list.Count; j++)
            {
               if (i == j) continue;
               var a = list[i];
               var t = list[j];
               if (!Side.CanMerge(b.Pieces[a.c, a.r], b.Pieces[t.c, t.r])) continue;
               var cov = _coverage[(int)b.Pieces[a.c, a.r].Kind];
               if (cov[a.c, a.r] > cov[t.c, t.r]) { var tmp = a; a = t; t = tmp; }
               return _s.CellToCell(a.c, a.r, t.c, t.r);
            }
         return false;
      }
   }
}
// ===== FILE: Assets/Scripts/Sim/V2.cs =====
using System;
namespace Rescue.Sim
{
   public struct V2
   {
      public float x, y;
      public V2(float x, float y) { this.x = x; this.y = y; }
      public static V2 operator +(V2 a, V2 b) => new V2(a.x + b.x, a.y + b.y);
      public static V2 operator -(V2 a, V2 b) => new V2(a.x - b.x, a.y - b.y);
      public static V2 operator *(V2 a, float k) => new V2(a.x * k, a.y * k);
      public float Len => (float)Math.Sqrt(x * x + y * y);
      public V2 Norm { get { var l = Len; return l < 1e-5f ? new V2(0, 0) : new V2(x / l, y / l...
      public static float Dist(V2 a, V2 b) => (a - b).Len;
      public static float Dot(V2 a, V2 b) => a.x * b.x + a.y * b.y;
```

```
13602       public static V2 Lerp(V2 a, V2 b, float t) => new V2(a.x + (b.x - a.x) * t, a.y + (b.y - ...
13603   }
13604   public sealed class Rng
13605   {
13606       uint _s;
13607       public Rng(uint seed) { _s = seed == 0 ? 0x9E3779B9u : seed; }
13608       public uint NextU()
13609       {
13610           _s ^= _s << 13; _s ^= _s >> 17; _s ^= _s << 5;
13611           return _s;
13612       }
13613       public float Next01() => (NextU() & 0xFFFFFF) / (float)0x1000000;
13614       public int Range(int minInclusive, int maxExclusive) => minInclusive + (int)(NextU() % (u...
13615       public bool Chance(float p) => Next01() < p;
13616   }
13617 }
13618 // ===== FILE: Assets/Scripts/Sim/Weapons.cs =====
13619 using System;
13620 using System.Collections.Generic;
13621 namespace Rescue.Sim
13622 {
13623   public enum WeaponKind { None, Blade, Spear, Bow, Axe, Club, Whip, Hammer, Fork, Sword, Stone...
13624   /// <summary>兵器按种类带一条特性，装备后在伤害倍率之外再生效。</summary>
13625   public struct WeaponAffix
13626   {
13627       public float Dmg, Range, Splash, Knock, Stun, Water, Crit, Cd, UltMul, UltCd;
13628       public int Pierce;
13629       public static readonly WeaponAffix Bare = new WeaponAffix { Dmg = 1f, Water = 1f, Cd = 1f...
13630   }
13631   public static class Weapons
13632   {
13633       public static readonly string[] Grades = { "凡器", "利器", "宝器", "神兵" };
13634       public static string KindName(WeaponKind k) => k switch
13635       {
13636           WeaponKind.Blade => "刀",
13637           WeaponKind.Spear => "枪",
13638           WeaponKind.Bow => "弓弩",
13639           WeaponKind.Axe => "斧",
13640           WeaponKind.Club => "棍棒",
13641           WeaponKind.Whip => "鞭锏",
13642           WeaponKind.Hammer => "锤",
13643           WeaponKind.Fork => "叉刺",
13644           WeaponKind.Sword => "剑",
13645           WeaponKind.Stone => "暗器",
13646           WeaponKind.Rope => "索",
13647           WeaponKind.Staff => "杖",
13648           WeaponKind.Halberd => "戟",
13649           WeaponKind.Cannon => "火炮",
13650           WeaponKind.Relic => "器物",
13651           WeaponKind.Fist => "拳脚",
```

```
13652         _ => "",
13653     };
13654     public static string Icon(WeaponKind k) => k switch
13655     {
13656         WeaponKind.Blade => "wp_blade",
13657         WeaponKind.Spear => "wp_spear",
13658         WeaponKind.Bow => "wp_bow",
13659         WeaponKind.Axe => "wp_axe",
13660         WeaponKind.Club => "wp_club",
13661         WeaponKind.Whip => "wp_whip",
13662         WeaponKind.Hammer => "wp_hammer",
13663         WeaponKind.Fork => "wp_fork",
13664         WeaponKind.Sword => "wp_sword",
13665         WeaponKind.Stone => "wp_stone",
13666         WeaponKind.Rope => "wp_rope",
13667         WeaponKind.Staff => "wp_staff",
13668         WeaponKind.Halberd => "wp_halberd",
13669         WeaponKind.Cannon => "wp_cannon",
13670         WeaponKind.Relic => "wp_fan",
13671         WeaponKind.Fist => "wp_fist",
13672         _ => "wp_blade",
13673     };
13674     public static string AffixText(WeaponKind k) => k switch
13675     {
13676         WeaponKind.Blade => "刀沉力猛：伤害再 +10%",
13677         WeaponKind.Spear => "长枪及远：多穿透一人（不穿透的射程 +0.3）",
13678         WeaponKind.Bow => "弓劲弦满：射程 +0.4",
13679         WeaponKind.Axe => "开山之势：溅射范围 +40%",
13680         WeaponKind.Club => "棍扫一片：命中击退小兵",
13681         WeaponKind.Whip => "鞭锏砸骨：命中定身 0.3 秒",
13682         WeaponKind.Hammer => "重锤千钧：命中定身 0.2 秒并击退",
13683         WeaponKind.Fork => "分水破浪：岸上 +8%，水上合计 ×1.4",
13684         WeaponKind.Sword => "剑走偏锋：暴击率 +12%",
13685         WeaponKind.Stone => "暗器连珠：出手速度 +12%",
13686         WeaponKind.Rope => "套索缠身：命中定身再 +0.4 秒",
13687         WeaponKind.Staff => "杖法通神：大招伤害 +30%",
13688         WeaponKind.Halberd => "戟锋横扫：溅射 +0.5 格",
13689         WeaponKind.Cannon => "火药充足：大招冷却 -15%",
13690         WeaponKind.Relic => "随身宝物：大招冷却 -20%",
13691         WeaponKind.Fist => "拳快如风：出手速度 +15%",
13692         _ => "",
13693     };
13694     public static WeaponAffix Affix(HeroDef d)
13695     {
13696         var a = WeaponAffix.Bare;
13697         switch (d.WKind)
13698         {
13699             case WeaponKind.Blade: a.Dmg = 1.1f; break;
13700             case WeaponKind.Spear: if (d.Pierce > 1) a.Pierce = 1; else a.Range = 0.3f; break;
13701             case WeaponKind.Bow: a.Range = 0.4f; break;
```

```
13702            case WeaponKind.Axe: a.Splash = d.Splash > 0 ? d.Splash * 0.4f : 0.5f; break;
13703            case WeaponKind.Club: a.Knock = 0.25f; break;
13704            case WeaponKind.Whip: a.Stun = 0.3f; break;
13705            case WeaponKind.Hammer: a.Stun = 0.2f; a.Knock = 0.2f; break;
13706            case WeaponKind.Fork: a.Water = 1.3f; a.Dmg = 1.08f; break;
13707            case WeaponKind.Sword: a.Crit = 0.12f; break;
13708            case WeaponKind.Stone: a.Cd = 0.88f; break;
13709            case WeaponKind.Rope: a.Stun = 0.4f; break;
13710            case WeaponKind.Staff: a.UltMul = 1.3f; break;
13711            case WeaponKind.Halberd: a.Splash = 0.5f; break;
13712            case WeaponKind.Cannon: a.UltCd = 0.85f; break;
13713            case WeaponKind.Relic: a.UltCd = 0.8f; break;
13714            case WeaponKind.Fist: a.Cd = 0.85f; break;
13715        }
13716        return a;
13717    }
13718    /// <summary>掉落权重：神兵最难得。</summary>
13719    public static int DropWeight(int grade) => grade switch { 3 => 1, 2 => 2, 1 => 3, _ => 4 };
13720    /// <summary>
13721    /// 缴获谁的兵器：先看这局凑出来的、已经入伙的；一个都没有，才从这一回里已经入伙的人里抽。
13722    /// 还没入伙的不掉，所以没过关之前，这一回才出场的人兵器不会亮。
13723    /// </summary>
13724    public static void CollectDropCandidates(IList<HeroId> formed, IList<HeroId> pool, Func<H...
13725    {
13726        if (formed != null)
13727            foreach (var id in formed)
13728                if (joined(id) && !string.IsNullOrEmpty(HeroDef.Get(id).Weapon)) into.Add(id);
13729        if (into.Count > 0 || pool == null) return;
13730        foreach (var id in pool)
13731            if (joined(id) && !string.IsNullOrEmpty(HeroDef.Get(id).Weapon)) into.Add(id);
13732    }
13733}
13734/// <summary>羁绊：同时凑出若干名有渊源的好汉，他们伤害提高。</summary>
13735public sealed class Bond
13736{
13737    public string Name, Story;
13738    public HeroId[] Members;
13739    public int Need;
13740    public float Dmg;
13741    public static readonly Bond[] All =
13742    {
13743        new Bond { Name = "阮氏三雄", Need = 2, Dmg = 0.25f, Story = "石碣村打鱼的三兄弟，同心智取生辰纲",
13744            Members = new[] { HeroId.RuanXiaoEr, HeroId.RuanXiaoWu, HeroId.RuanXiaoQi } },
13745        new Bond { Name = "五虎上将", Need = 2, Dmg = 0.15f, Story = "梁山马军五虎将",
13746            Members = new[] { HeroId.GuanSheng, HeroId.LinChong, HeroId.QinMing, HeroId.HuYan...
13747        new Bond { Name = "八骠骑", Need = 2, Dmg = 0.12f, Story = "马军八骠骑兼先锋使",
13748            Members = new[] { HeroId.HuaRong, HeroId.XuNing, HeroId.YangZhi, HeroId.SuoChao, ...
13749        new Bond { Name = "二龙山", Need = 2, Dmg = 0.15f, Story = "鲁智深、杨志、武松同占二龙山",
13750            Members = new[] { HeroId.LuZhiShen, HeroId.YangZhi, HeroId.WuSong, HeroId.CaoZhen...
13751        new Bond { Name = "七星聚义", Need = 3, Dmg = 0.15f, Story = "晁盖与七人聚义，智取生辰纲",
```

```
            Members = new[] { HeroId.WuYong, HeroId.GongSunSheng, HeroId.LiuTang, HeroId.Ruan...
        new Bond { Name = "张氏兄弟", Need = 2, Dmg = 0.2f, Story = "浔阳江上船火儿与浪里白条",
            Members = new[] { HeroId.ZhangHeng, HeroId.ZhangShun } },
        new Bond { Name = "解氏兄弟", Need = 2, Dmg = 0.2f, Story = "登州打虎的两头蛇与双尾蝎",
            Members = new[] { HeroId.XieZhen, HeroId.XieBao } },
        new Bond { Name = "翠屏山", Need = 2, Dmg = 0.18f, Story = "杨雄、石秀结为兄弟，带时迁投梁山",
            Members = new[] { HeroId.YangXiong, HeroId.ShiXiu, HeroId.ShiQian } },
        new Bond { Name = "少华山", Need = 2, Dmg = 0.15f, Story = "史进与朱武、陈达、杨春不打不相识",
            Members = new[] { HeroId.ShiJin, HeroId.ZhuWu, HeroId.ChenDa, HeroId.YangChun } },
        new Bond { Name = "十字坡夫妻", Need = 2, Dmg = 0.18f, Story = "张青、孙二娘夫妻开店",
            Members = new[] { HeroId.ZhangQingY, HeroId.SunErNiang } },
        new Bond { Name = "王英扈三娘", Need = 2, Dmg = 0.18f, Story = "宋江作主，一丈青嫁了矮脚虎",
            Members = new[] { HeroId.WangYing, HeroId.HuSanNiang } },
        new Bond { Name = "登州孙家", Need = 2, Dmg = 0.15f, Story = "孙立、孙新、顾大嫂登州劫牢",
            Members = new[] { HeroId.SunLi, HeroId.SunXin, HeroId.GuDaSao, HeroId.YueHe } },
        new Bond { Name = "芒砀三杰", Need = 2, Dmg = 0.18f, Story = "混世魔王樊瑞与左右手项充、李衮",
            Members = new[] { HeroId.FanRui, HeroId.XiangChong, HeroId.LiGun } },
        new Bond { Name = "画戟双英", Need = 2, Dmg = 0.2f, Story = "吕方、郭盛对影山斗戟，花荣一箭射开",
            Members = new[] { HeroId.LvFang, HeroId.GuoSheng } },
        new Bond { Name = "揭阳水军", Need = 2, Dmg = 0.15f, Story = "李俊与童威、童猛浔阳江上做私商",
            Members = new[] { HeroId.LiJun, HeroId.TongWei, HeroId.TongMeng, HeroId.LiLi } },
        new Bond { Name = "铁牛与哥哥", Need = 2, Dmg = 0.2f, Story = "黑旋风生死追随宋江",
            Members = new[] { HeroId.SongJiang, HeroId.LiKui } },
        new Bond { Name = "主仆情深", Need = 2, Dmg = 0.2f, Story = "浪子燕青舍命救主卢俊义",
            Members = new[] { HeroId.LuJunYi, HeroId.YanQing } },
        new Bond { Name = "郓城都头", Need = 2, Dmg = 0.18f, Story = "朱仝、雷横先后私放晁盖、宋江",
            Members = new[] { HeroId.ZhuTong, HeroId.LeiHeng } },
        new Bond { Name = "水火二将", Need = 2, Dmg = 0.2f, Story = "凌州圣水将、神火将",
            Members = new[] { HeroId.ShanTingGui, HeroId.WeiDingGuo } },
        new Bond { Name = "江州好汉", Need = 2, Dmg = 0.12f, Story = "江州劫法场救宋江",
            Members = new[] { HeroId.DaiZong, HeroId.LiKui, HeroId.ZhangShun, HeroId.LiJun, H...
    };
  }
}
// ===== FILE: cloudfunctions/songjiang-api/index.js =====
/**
 * 营救宋江后端（CloudBase 事件云函数 + HTTP 访问），照 huahua-api。
 * gameKey 与经分一致：songjiang。
 *
 *   POST /login
 *   POST /save/pull
 *   POST /save/push
 *   GET|POST /health
 *
 * 环境变量：GAME_KEY=songjiang，SONGJIANG_JWT_SECRET 必填。
 * 微信真机再配 SONGJIANG_WX_APPID / SONGJIANG_WX_SECRET。
 * 集合：songjiang_playerData（userId 唯一索引）
 */
const { handleLogin } = require('./lib/auth');
const { handlePull, handlePush } = require('./lib/save');
```

```
const { respond, parseEvent, preflight } = require('./lib/http');
const ROUTES = {
  'GET /health': async () => ({ ok: true, ts: Date.now() }),
  'POST /health': async () => ({ ok: true, ts: Date.now() }),
  'POST /login': handleLogin,
  'POST /save/pull': handlePull,
  'POST /save/push': handlePush,
};
exports.main = async (event, context) => {
  try {
    if (event && event.httpMethod === 'OPTIONS') return preflight();
    const req = parseEvent(event);
    const key = `${req.method} ${req.path}`;
    const handler = ROUTES[key];
    if (!handler) {
      return respond(404, { ok: false, code: 'NOT_FOUND', error: `no route: ${key}` });
    }
    const result = await handler(req, context);
    if (result && typeof result === 'object' && 'statusCode' in result) return result;
    return respond(200, { ok: true, data: result });
  } catch (error) {
    const code = error && error.code ? error.code : 'INTERNAL';
    const status = error && error.status ? error.status : 500;
    const message = (error && error.message) || String(error);
    console.error('[songjiang-api] error:', code, message, error && error.stack);
    const out = { ok: false, code, error: message };
    if (error && error.data !== undefined) out.data = error.data;
    return respond(status, out);
  }
};
// ===== FILE: cloudfunctions/songjiang-api/lib/auth.js =====
const jwt = require('jsonwebtoken');
const { httpError } = require('./http');
const {
  getGameKey,
  gameKeyUpper,
  getJwtSecret: readJwtSecret,
  getTtlSec,
  getPlatformCredential,
} = require('./config');
const SUPPORTED_PLATFORMS = new Set(['wx', 'dy', 'anon']);
function getJwtSecret() {
  const secret = readJwtSecret();
  if (!secret) {
    throw httpError(500, 'NO_JWT_SECRET', `${gameKeyUpper()}_JWT_SECRET 未配置`);
  }
  return secret;
}
async function handleLogin(req) {
  const body = req.body || {};
```

```
  const platform = String(body.platform || '').toLowerCase();
  if (!SUPPORTED_PLATFORMS.has(platform)) {
    throw httpError(400, 'BAD_PLATFORM', `unsupported platform: ${platform}`);
  }
  let platformUid = '';
  if (platform === 'wx') {
    platformUid = (await wxCode2Openid(body.code)).openid;
  } else if (platform === 'dy') {
    platformUid = await ttCode2Openid(body.code);
  } else if (platform === 'anon') {
    const id = String(body.anonId || '').trim();
    if (!id) throw httpError(400, 'NO_ANON_ID', 'anonId 缺失');
    if (!/^[A-Za-z0-9_\-:.]{8,128}$/.test(id)) {
      throw httpError(400, 'BAD_ANON_ID', 'anonId 非法');
    }
    platformUid = id;
  }
  const userId = `${platform}:${platformUid}`;
  const ttlSec = getTtlSec();
  const gameKey = getGameKey();
  const now = Math.floor(Date.now() / 1000);
  const token = jwt.sign(
    { sub: userId, plt: platform, gk: gameKey, iat: now },
    getJwtSecret(),
    { expiresIn: ttlSec },
  );
  return {
    token,
    userId,
    platform,
    gameKey,
    expiresAt: (now + ttlSec) * 1000,
    ttlSec,
  };
}
function requireUser(req) {
  const authHeader = (req.headers && req.headers.authorization) || '';
  const match = /^Bearer\s+(.+)$/i.exec(authHeader);
  if (!match) throw httpError(401, 'NO_TOKEN', '缺少 Authorization: Bearer <token>');
  let payload;
  try {
    payload = jwt.verify(match[1].trim(), getJwtSecret());
  } catch (error) {
    throw httpError(401, 'BAD_TOKEN', error && error.message ? error.message : 'token 无效');
  }
  const userId = payload && payload.sub;
  if (!userId || typeof userId !== 'string' || !userId.includes(':')) {
    throw httpError(401, 'BAD_TOKEN', 'token sub 非法');
  }
  const currentGameKey = getGameKey();
```

```
13902   if (payload.gk && payload.gk !== currentGameKey) {
13903     throw httpError(401, 'BAD_TOKEN', `token gameKey=${payload.gk} 与当前 GAME_KEY=${currentGameKey}...
13904   }
13905   return { userId, platform: payload.plt || userId.split(':')[0] };
13906 }
13907 async function wxCode2Openid(code) {
13908   const appid = getPlatformCredential('wx', 'APPID');
13909   const secret = getPlatformCredential('wx', 'SECRET');
13910   if (!appid || !secret) throw httpError(500, 'NO_WX_CFG', `${gameKeyUpper()}_WX_APPID/${gameKeyU...
13911   if (!code) throw httpError(400, 'NO_CODE', 'wx code 缺失');
13912   const url = `https://api.weixin.qq.com/sns/jscode2session?appid=${encodeURIComponent(appid)}&se...
13913   const data = await httpGetJson(url);
13914   if (!data || !data.openid) {
13915     throw httpError(401, 'WX_LOGIN_FAIL', `wx code2session 失败: ${JSON.stringify(data || {})}`);
13916   }
13917   return { openid: data.openid, sessionKey: data.session_key || '' };
13918 }
13919 async function ttCode2Openid(code) {
13920   const appid = getPlatformCredential('tt', 'APPID');
13921   const secret = getPlatformCredential('tt', 'SECRET');
13922   if (!appid || !secret) throw httpError(500, 'NO_TT_CFG', `${gameKeyUpper()}_TT_APPID/${gameKeyU...
13923   if (!code) throw httpError(400, 'NO_CODE', 'dy code 缺失');
13924   const data = await httpPostJson('https://developer.toutiao.com/api/apps/v2/jscode2session', { a...
13925   if (!data || data.err_no !== 0 || !data.data || !data.data.openid) {
13926     throw httpError(401, 'TT_LOGIN_FAIL', `dy code2session 失败: ${JSON.stringify(data || {})}`);
13927   }
13928   return data.data.openid;
13929 }
13930 function httpGetJson(url) {
13931   return httpRequestJson(url, 'GET');
13932 }
13933 function httpPostJson(url, body) {
13934   return httpRequestJson(url, 'POST', body);
13935 }
13936 function httpRequestJson(url, method, body) {
13937   if (typeof fetch !== 'function') {
13938     return Promise.reject(httpError(500, 'NO_FETCH', '当前 Node 运行时不支持 fetch，请使用 Node 18+'));
13939   }
13940   const opts = { method, headers: { 'Content-Type': 'application/json' } };
13941   if (body !== undefined) opts.body = JSON.stringify(body);
13942   return fetch(url, opts).then(async (res) => {
13943     const text = await res.text();
13944     try {
13945       return JSON.parse(text);
13946     } catch (_) {
13947       return { _raw: text };
13948     }
13949   });
13950 }
13951 module.exports = {
```

```
  handleLogin,
  requireUser,
};
// ===== FILE: cloudfunctions/songjiang-api/lib/config.js =====
const DEFAULT_GAME_KEY = 'songjiang';
const DEFAULT_TTL_SEC = 7 * 24 * 3600;
const DEFAULT_MAX_BYTES = 256 * 1024;
function getGameKey() {
  const v = String(process.env.GAME_KEY || '').trim().toLowerCase();
  if (!v) return DEFAULT_GAME_KEY;
  if (!/^[a-z][a-z0-9_\-]{0,31}$/.test(v)) {
    throw new Error(`非法 GAME_KEY: ${v}`);
  }
  return v;
}
function gameKeyUpper() {
  return getGameKey().toUpperCase().replace(/[^A-Z0-9]/g, '_');
}
function readEnvPrefer(...keys) {
  for (const key of keys) {
    const value = process.env[key];
    if (value !== undefined && value !== null && String(value).length > 0) {
      return String(value);
    }
  }
  return '';
}
function getCollectionName(suffix) {
  const overrideKey = `${gameKeyUpper()}_${suffix.toUpperCase()}_COLLECTION`;
  const override = process.env[overrideKey];
  if (override) return String(override);
  return `${getGameKey()}_${suffix}`;
}
function getJwtSecret() {
  return readEnvPrefer(`${gameKeyUpper()}_JWT_SECRET`);
}
function getTtlSec() {
  const raw = readEnvPrefer(`${gameKeyUpper()}_TOKEN_TTL_SEC`);
  const v = Number(raw);
  return Number.isFinite(v) && v > 0 ? Math.floor(v) : DEFAULT_TTL_SEC;
}
function getMaxBytes() {
  const raw = readEnvPrefer(`${gameKeyUpper()}_SAVE_MAX_BYTES`);
  const v = Number(raw);
  return Number.isFinite(v) && v > 0 ? Math.floor(v) : DEFAULT_MAX_BYTES;
}
function getPlatformCredential(platform, field) {
  const upper = gameKeyUpper();
  const platformUpper = platform.toUpperCase();
  return readEnvPrefer(`${upper}_${platformUpper}_${field}`);
```

```
}
module.exports = {
  getGameKey,
  gameKeyUpper,
  readEnvPrefer,
  getCollectionName,
  getJwtSecret,
  getTtlSec,
  getMaxBytes,
  getPlatformCredential,
};
// ===== FILE: cloudfunctions/songjiang-api/lib/db.js =====
const tcb = require('@cloudbase/node-sdk');
const { getCollectionName } = require('./config');
let app = null;
function getApp() {
  if (app) return app;
  app = tcb.init({
    env: process.env.TCB_ENV || tcb.SYMBOL_CURRENT_ENV,
  });
  return app;
}
function getCollection() {
  return getApp().database().collection(getCollectionName('playerData'));
}
module.exports = {
  getApp,
  getCollection,
};
// ===== FILE: cloudfunctions/songjiang-api/lib/http.js =====
const { getGameKey } = require('./config');
const CORS_HEADERS = {
  'Access-Control-Allow-Origin': '*',
  'Access-Control-Allow-Methods': 'GET,POST,OPTIONS',
  'Access-Control-Allow-Headers': 'Content-Type,Authorization',
  'Access-Control-Max-Age': '86400',
};
function respond(statusCode, body, extraHeaders = {}) {
  return {
    statusCode,
    headers: {
      'Content-Type': 'application/json; charset=utf-8',
      ...CORS_HEADERS,
      ...extraHeaders,
    },
    body: typeof body === 'string' ? body : JSON.stringify(body),
    isBase64Encoded: false,
  };
}
function preflight() {
```

```
  return {
    statusCode: 204,
    headers: { ...CORS_HEADERS },
    body: '',
    isBase64Encoded: false,
  };
}
function parseEvent(event) {
  event = event || {};
  if (event.httpMethod) {
    const method = String(event.httpMethod).toUpperCase();
    let rawBody = event.body || '';
    if (event.isBase64Encoded && rawBody) {
      try {
        rawBody = Buffer.from(rawBody, 'base64').toString('utf8');
      } catch (_) {}
    }
    let body = {};
    if (rawBody) {
      try {
        body = JSON.parse(rawBody);
      } catch (_) {
        body = {};
      }
    }
    return {
      method,
      path: normalizePath(event.path || '/'),
      body,
      headers: lowercaseHeaders(event.headers || {}),
      query: event.queryStringParameters || {},
      raw: event,
    };
  }
  const action = String(event.action || '').replace(/^\/+/, '');
  return {
    method: 'POST',
    path: action ? `/${action}` : '/',
    body: event.body || {},
    headers: lowercaseHeaders(event.headers || {}),
    query: {},
    raw: event,
  };
}
function normalizePath(path) {
  if (!path) return '/';
  let p = String(path);
  if (!p.startsWith('/')) p = `/${p}`;
  const apiPrefix = `${getGameKey()}-api`;
  p = p.replace(new RegExp(`^/(?:${escapeRegExp(apiPrefix)})(?=/|$)`), '');
```

```
  if (p === '') p = '/';
  if (p.length > 1 && p.endsWith('/')) p = p.slice(0, -1);
  return p;
}
function lowercaseHeaders(headers) {
  const out = {};
  for (const key of Object.keys(headers || {})) {
    out[key.toLowerCase()] = headers[key];
  }
  return out;
}
function httpError(status, code, message) {
  const err = new Error(message || code);
  err.status = status;
  err.code = code;
  return err;
}
function escapeRegExp(text) {
  return String(text).replace(/[.*+?^${}()|[\]\\]/g, '\\$&');
}
module.exports = {
  respond,
  preflight,
  parseEvent,
  httpError,
};
// ===== FILE: cloudfunctions/songjiang-api/lib/save.js =====
const { httpError } = require('./http');
const { requireUser } = require('./auth');
const { getCollection } = require('./db');
const { getMaxBytes } = require('./config');
async function handlePull(req) {
  const { userId, platform } = requireUser(req);
  const col = getCollection();
  const res = await col.where({ userId }).limit(1).get();
  const doc = (res && Array.isArray(res.data) && res.data[0]) || null;
  if (!doc) {
    return {
      userId,
      platform,
      exists: false,
      schemaVersion: 0,
      updatedAt: 0,
      payload: {},
      payloadKeys: [],
    };
  }
  return {
    userId,
    platform,
```

```
    exists: true,
    schemaVersion: doc.schemaVersion || 0,
    updatedAt: doc.updatedAt || 0,
    payload: doc.payload || {},
    payloadKeys: Array.isArray(doc.payloadKeys) ? doc.payloadKeys : Object.keys(doc.payload || {}),
    clientFingerprint: doc.clientFingerprint || '',
  };
}
async function handlePush(req) {
  const { userId, platform } = requireUser(req);
  const body = req.body || {};
  const schemaVersion = Number(body.schemaVersion);
  const updatedAt = Number(body.updatedAt);
  const baseRemoteUpdatedAt = Number(body.baseRemoteUpdatedAt || 0);
  const payload = body.payload;
  const clientFingerprint = String(body.clientFingerprint || '').slice(0, 200);
  if (!Number.isFinite(schemaVersion) || schemaVersion <= 0) {
    throw httpError(400, 'BAD_SCHEMA', 'schemaVersion 非法');
  }
  if (!Number.isFinite(updatedAt) || updatedAt <= 0) {
    throw httpError(400, 'BAD_UPDATED_AT', 'updatedAt 非法');
  }
  if (!Number.isFinite(baseRemoteUpdatedAt) || baseRemoteUpdatedAt < 0) {
    throw httpError(400, 'BAD_BASE_REMOTE_UPDATED_AT', 'baseRemoteUpdatedAt 非法');
  }
  if (!payload || typeof payload !== 'object' || Array.isArray(payload)) {
    throw httpError(400, 'BAD_PAYLOAD', 'payload 必须是 object');
  }
  const size = Buffer.byteLength(JSON.stringify(payload), 'utf8');
  const maxBytes = getMaxBytes();
  if (size > maxBytes) {
    throw httpError(413, 'PAYLOAD_TOO_LARGE', `payload 超限: ${size}B > ${maxBytes}B`);
  }
  const payloadKeys = [];
  for (const k of Object.keys(payload)) {
    if (typeof payload[k] !== 'string') {
      throw httpError(400, 'BAD_PAYLOAD_VALUE', `payload[${k}] 必须是字符串`);
    }
    payloadKeys.push(k);
  }
  const col = getCollection();
  const existingRes = await col.where({ userId }).limit(1).get();
  const existing = (existingRes && Array.isArray(existingRes.data) && existingRes.data[0]) || null;
  if (existing) {
    const prevUpdatedAt = Number(existing.updatedAt) || 0;
    if (updatedAt < prevUpdatedAt || baseRemoteUpdatedAt < prevUpdatedAt) {
      throw Object.assign(
        httpError(409, 'STALE_UPDATE', `服务端已有更新版本 remote=${prevUpdatedAt} > local=${updatedAt}, b...
        {
          data: {
```

```
          remote: {
            schemaVersion: existing.schemaVersion || 0,
            updatedAt: prevUpdatedAt,
            payload: existing.payload || {},
            payloadKeys: Array.isArray(existing.payloadKeys)
              ? existing.payloadKeys
              : Object.keys(existing.payload || {}),
          },
        },
      },
    );
  }
}
const mergedPayload = { ...(existing && existing.payload ? existing.payload : {}) };
for (const k of Object.keys(payload)) mergedPayload[k] = payload[k];
const mergedPayloadKeys = Object.keys(mergedPayload);
const mergedSize = Buffer.byteLength(JSON.stringify(mergedPayload), 'utf8');
if (mergedSize > maxBytes) {
  throw httpError(413, 'PAYLOAD_TOO_LARGE', `payload 超限: ${mergedSize}B > ${maxBytes}B`);
}
const now = Date.now();
const docData = {
  userId,
  platform,
  schemaVersion,
  updatedAt,
  baseRemoteUpdatedAt,
  clientFingerprint,
  payload: mergedPayload,
  payloadKeys: mergedPayloadKeys,
  lastWriteAt: now,
};
if (existing && existing._id) {
  await col.doc(existing._id).update(docData);
  return { userId, updatedAt, savedAt: now, mode: 'update', sizeBytes: mergedSize };
}
const addRes = await col.add(docData);
return {
  userId,
  updatedAt,
  savedAt: now,
  mode: 'insert',
  sizeBytes: size,
  _id: addRes && (addRes.id || addRes._id),
};
}
module.exports = {
  handlePull,
  handlePush,
};
```